
SMS Consolidated Implementation Verification Checklist

Addendums 31, 32, 33 & 34

Document: SMS-IMPL-VERIFY-CONSOLIDATED v1.0

Covers: SMS-FSD-ADD-031, SMS-FSD-ADD-032, SMS-FSD-ADD-033, SMS-FSD-ADD-034

Purpose: Manual verification of all implementation steps across four addendums

Date: 2026-10-06

Total Changes: 26 changes across 4 addendums

Total Migrations: 18 schema migrations

Total Business Rules: 107 business rules

Total Test Scenarios: 128 test scenarios

Permission Claims: 14 new permission claims

Dev Timeline: ~76 days (parallel) / 90 days (sequential)

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PART 1 — SCHEMA MIGRATIONS (18 Total)

1.1 Addendum 31 Migrations (M1–M3)

M1: Product Min/Max Qty Columns

Table: `lookups.Products`

```
ALTER TABLE lookups.Products
  ADD sale_order_min_qty DECIMAL(18,4) NULL;

ALTER TABLE lookups.Products
  ADD sale_order_max_qty DECIMAL(18,4) NULL;
```

- Column `sale_order_min_qty` exists on `lookups.Products` as `DECIMAL(18,4) NULL`
- Column `sale_order_max_qty` exists on `lookups.Products` as `DECIMAL(18,4) NULL`
- Existing product rows unaffected (NULL values)
- EF Core entity `Product` updated with both properties

M2: Production Order ↔ Sale Order Link

Table: `material.ProductionOrders`

```
ALTER TABLE material.ProductionOrders
  ADD sale_order_id BIGINT NULL
  CONSTRAINT FK_ProductionOrders_SaleOrders
  FOREIGN KEY REFERENCES demand.SaleOrders(sale_order_id);

ALTER TABLE material.ProductionOrders
  ADD sale_order_line_id BIGINT NULL
  CONSTRAINT FK_ProductionOrders_SaleOrderLines
  FOREIGN KEY REFERENCES demand.SaleOrderLines(sale_order_line_id);

CREATE INDEX IX_ProductionOrders_SaleOrder
  ON material.ProductionOrders(sale_order_id)
  WHERE sale_order_id IS NOT NULL;
```

- Column `sale_order_id` exists as `BIGINT NULL` with FK to `demand.SaleOrders`
- Column `sale_order_line_id` exists as `BIGINT NULL` with FK to `demand.SaleOrderLines`
- Filtered index `IX_ProductionOrders_SaleOrder` exists on `sale_order_id` WHERE `sale_order_id IS NOT NULL`
- EF Core entity `ProductionOrder` has navigation properties to `SaleOrder` and `SaleOrderLine`
- Existing production orders unaffected (NULL values)

M3: Remove Warehouse from BOM Lines

Table: `material.BillofMaterialLines`

```
ALTER TABLE material.BillofMaterialLines
  DROP CONSTRAINT IF EXISTS FK_BOMLines_Warehouse;

ALTER TABLE material.BillofMaterialLines
  DROP COLUMN IF EXISTS warehouse_id;
```

- Column `warehouse_id` no longer exists on `material.BillofMaterialLines`
- FK constraint `FK_BOMLines_Warehouse` dropped
- EF Core entity `BillofMaterialLine` no longer has `WarehouseId` property
- All code references to `BOMLine.warehouse_id` removed
- PMR warehouse now always sourced from `ProductionOrder.production_warehouse_id`

1.2 Addendum 32 Migrations (M1–M7)

M1: Rejection Reasons Lookup

Table: `lookups.RejectionReasons`

```
CREATE TABLE lookups.RejectionReasons (
    rejection_reason_id INT IDENTITY(1,1) PRIMARY KEY,
    org_id INT NOT NULL REFERENCES auth.Organizations(org_id),
    code NVARCHAR(10) NOT NULL,
    name NVARCHAR(100) NOT NULL,
    description NVARCHAR(500) NULL,
    is_active BIT NOT NULL DEFAULT 1,
    is_system BIT NOT NULL DEFAULT 0,
    created_by INT NOT NULL,
    created_at DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    modified_by INT NULL,
    modified_at DATETIME2 NULL,
    CONSTRAINT UQ_RejectionReasons_Code UNIQUE(org_id, code)
);
```

Seed Data (10 codes):

Code	Name
NIP	Not in Product Range
OOS	Out of Stock
DIS	Discontinued
MOQ	Below Minimum Order Quantity
GEO	Geographic Restriction
REG	Regulatory Restriction
CAP	Capacity Constraint
CRD	Credit Hold
PRC	Price Disagreement
OTH	Other

- Table `lookups.RejectionReasons` created with all columns
- Unique constraint on `(org_id, code)` exists
- All 10 seed codes inserted for each existing org
- `is_system = 1` for all seed records (prevents deletion)
- EF Core entity and `HasQueryFilter(r => r.OrgId == _orgId)` configured

M2: Sale Inquiry Tables

Tables: `demand.SaleInquiries`, `demand.SaleInquiryLines`, `demand.SaleInquiryAttachments`

- Table `demand.SaleInquiries` created with columns:
 - `inquiry_id` BIGINT IDENTITY PK
 - `org_id` INT NOT NULL FK
 - `inquiry_number` NVARCHAR(20) NOT NULL (format: `INQ-{YYYY}-{NNNN}`)
 - `customer_id` INT NOT NULL FK to `suppliers.Suppliers`
 - `inquiry_date` DATE NOT NULL
 - `status` NVARCHAR(20) NOT NULL DEFAULT 'RECEIVED'
 - `notes` NVARCHAR(2000) NULL
 - `assigned_to` INT NULL FK to `auth.Users`
 - Standard audit columns (`created_by`, `created_at`, `modified_by`, `modified_at`)
 - Unique constraint on `(org_id, inquiry_number)`
- Table `demand.SaleInquiryLines` created with columns:

- `inquiry_line_id` BIGINT IDENTITY PK
- `inquiry_id` BIGINT NOT NULL FK
- `product_variant_id` INT NOT NULL FK
- `requested_qty` DECIMAL(18,4) NOT NULL
- `unit_id` INT NOT NULL FK
- `requested_delivery_date` DATE NULL
- `evaluation_status` NVARCHAR(20) NOT NULL DEFAULT 'PENDING'
- `evaluation_notes` NVARCHAR(1000) NULL
- `rejection_reason_id` INT NULL FK
- `available_qty` DECIMAL(18,4) NULL
- `earliest_delivery_date` DATE NULL
- `line_order` INT NOT NULL DEFAULT 0
- Table `demand.SaleInquiryAttachments` created
- Index on (`org_id`, `status`) exists on `SaleInquiries`
- Index on (`inquiry_id`) exists on `SaleInquiryLines`

M3: Sale Quotation Tables

Tables: `demand.SaleQuotations`, `demand.SaleQuotationLines`, `demand.SaleQuotationAttachments`

- Table `demand.SaleQuotations` created with columns:
 - `quotation_id` BIGINT IDENTITY PK
 - `org_id` INT NOT NULL FK
 - `quotation_number` NVARCHAR(20) NOT NULL (format: `SQ-{YYYYY}-{NNNN}`)
 - `customer_id` INT NOT NULL FK
 - `inquiry_id` BIGINT NULL FK (link back to inquiry)
 - `quotation_date` DATE NOT NULL
 - `valid_until` DATE NULL
 - `status` NVARCHAR(20) NOT NULL DEFAULT 'DRAFT'
 - `currency_id` INT NOT NULL FK
 - `notes` NVARCHAR(2000) NULL
 - Standard audit columns
 - Unique constraint on (`org_id`, `quotation_number`)
- Table `demand.SaleQuotationLines` created with columns:
 - `quotation_line_id` BIGINT IDENTITY PK
 - `quotation_id` BIGINT NOT NULL FK
 - `product_variant_id` INT NOT NULL FK
 - `qty` DECIMAL(18,4) NOT NULL
 - `unit_id` INT NOT NULL FK
 - `unit_price` DECIMAL(18,4) NOT NULL
 - `discount_percent` DECIMAL(5,2) NULL DEFAULT 0
 - `line_total` AS (`qty * unit_price * (1 - ISNULL(discount_percent,0)/100)`) PERSISTED
 - `line_type` NVARCHAR(20) NOT NULL DEFAULT 'NORMAL' (NORMAL/REJECTED/ALTERNATIVE)
 - `rejection_reason_id` INT NULL FK
 - `parent_line_id` BIGINT NULL FK (self-ref for alternatives)
 - `customer_response` NVARCHAR(20) NOT NULL DEFAULT 'PENDING' (PENDING/ACCEPTED/REJECTED/COUNTER)
 - `counter_price` DECIMAL(18,4) NULL
 - `counter_qty` DECIMAL(18,4) NULL
 - `delivery_date` DATE NULL
 - `line_order` INT NOT NULL DEFAULT 0
- Table `demand.SaleQuotationAttachments` created

- Index on (**org_id**, **status**) exists on SaleQuotations
- Index on (**quotation_id**, **line_type**) exists on SaleQuotationLines

M4: Sale Order Source Linking

Table: **demand.SaleOrders** (ALTER)

```
ALTER TABLE demand.SaleOrders ADD source_quotation_id BIGINT NULL
CONSTRAINT FK_SaleOrders_SourceQuotation
FOREIGN KEY REFERENCES demand.SaleQuotations(quotation_id);

ALTER TABLE demand.SaleOrders ADD source_inquiry_id BIGINT NULL
CONSTRAINT FK_SaleOrders_SourceInquiry
FOREIGN KEY REFERENCES demand.SaleInquiries(inquiry_id);

ALTER TABLE demand.SaleOrders ADD source_type NVARCHAR(20) NOT NULL
DEFAULT 'MANUAL'; -- MANUAL/FROM_QUOTATION/PORTAL/INTER_TENANT

ALTER TABLE demand.SaleOrders ADD customer_po_reference NVARCHAR(50) NULL;
ALTER TABLE demand.SaleOrders ADD customer_po_date DATE NULL;
ALTER TABLE demand.SaleOrders ADD customer_po_attachment_id BIGINT NULL;
```

- Column **source_quotation_id** exists as **BIGINT NULL** with FK
- Column **source_inquiry_id** exists as **BIGINT NULL** with FK
- Column **source_type** exists as **NVARCHAR(20) NOT NULL DEFAULT 'MANUAL'**
- Column **customer_po_reference** exists as **NVARCHAR(50) NULL**
- Column **customer_po_date** exists as **DATE NULL**
- Column **customer_po_attachment_id** exists as **BIGINT NULL**
- Existing sale orders backfilled with **source_type = 'MANUAL'**
- Index on **source_quotation_id** where not null
- Index on **source_inquiry_id** where not null

M5: Sale Order Attachments

Table: **demand.SaleOrderAttachments**

- Table **demand.SaleOrderAttachments** created with standard attachment columns
- FK to **demand.SaleOrders** exists
- **org_id** discriminator and query filter configured

M6: Sale Order Line Reserved Qty

Table: **demand.SaleOrderLines** (ALTER)

```
ALTER TABLE demand.SaleOrderLines
ADD reserved_qty DECIMAL(18,4) NOT NULL DEFAULT 0;

-- Backfill existing
UPDATE demand.SaleOrderLines SET reserved_qty = 0 WHERE reserved_qty = 0;
```

- Column **reserved_qty** exists as **DECIMAL(18,4) NOT NULL DEFAULT 0**
- All existing rows have **reserved_qty = 0**
- EF Core entity updated

M7: Sales Permission Claims

Table: **auth.Permissions**

```
INSERT INTO auth.Permissions (permission_name, description, module)
VALUES
('sales_inquiry_view', 'View sale inquiries', 'demand'),
('sales_inquiry_create', 'Create sale inquiries', 'demand'),
('sales_inquiry_edit', 'Edit sale inquiries', 'demand'),
('sales_quotation_view', 'View sale quotations', 'demand'),
('sales_quotation_create', 'Create sale quotations', 'demand');
```

```
( 'sales_quotation_edit', 'Edit sale quotations', 'demand'),
( 'sales_quotation_send', 'Send quotations to customers', 'demand'),
( 'sales_order_create', 'Create sale orders', 'demand'),
( 'sales_reserve_inventory', 'Reserve inventory against SO lines', 'demand'),
( 'sales_release_reservation', 'Release inventory reservations', 'demand');
```

- All 10 permission claims inserted
- Claims available in role assignment UI
- `[Authorize(Policy = "sales_inquiry_view")]` etc. applied to correct controllers

1.3 Addendum 33 Migrations (M1–M4)

M1: Fulfillment Route Configuration

Tables: `logistics.FulfillmentRoutes`, `logistics.FulfillmentRouteSteps`

```
CREATE TABLE logistics.FulfillmentRoutes (
    fulfillment_route_id INT IDENTITY(1,1) PRIMARY KEY,
    org_id INT NOT NULL REFERENCES auth.Organizations(org_id),
    code NVARCHAR(30) NOT NULL,
    name NVARCHAR(100) NOT NULL,
    description NVARCHAR(500) NULL,
    is_default BIT NOT NULL DEFAULT 0,
    is_active BIT NOT NULL DEFAULT 1,
    is_system BIT NOT NULL DEFAULT 0,
    requires_packing BIT NOT NULL DEFAULT 0,
    requires_shipping BIT NOT NULL DEFAULT 1,
    -- Standard audit columns
    CONSTRAINT UQ_FulfillmentRoutes_Code UNIQUE(org_id, code)
);

CREATE TABLE logistics.FulfillmentRouteSteps (
    route_step_id INT IDENTITY(1,1) PRIMARY KEY,
    fulfillment_route_id INT NOT NULL REFERENCES logistics.FulfillmentRoutes,
    step_code NVARCHAR(20) NOT NULL
    CHECK (step_code IN ('PICK','PACK','STAGE','APPROVAL','GOODS_ISSUE','SHIP')),
    step_order INT NOT NULL,
    is_mandatory BIT NOT NULL DEFAULT 1,
    CONSTRAINT UQ_RouteSteps_Order UNIQUE(fulfillment_route_id, step_order)
);
```

3 Seed Routes:

Code	Name	Default	Steps
PICK_ONLY	Pick Only	No	Pick → Goods Issue
PICK_AND_SHIP	Pick and Ship	Yes	Pick → Goods Issue → Ship
PICK_PACK_SHIP	Pick, Pack and Ship	No	Pick → Pack → Goods Issue → Ship

- Table `logistics.FulfillmentRoutes` created with all columns
- Table `logistics.FulfillmentRouteSteps` created with CHECK constraint on `step_code`
- Unique constraint on `(org_id, code)` exists
- Unique constraint on `(fulfillment_route_id, step_order)` exists
- Seed route `PICK_ONLY` inserted with steps: PICK(1) → GOODS_ISSUE(2)
- Seed route `PICK_AND_SHIP` inserted with steps: PICK(1) → GOODS_ISSUE(2) → SHIP(3), `is_default = 1`
- Seed route `PICK_PACK_SHIP` inserted with steps: PICK(1) → PACK(2) → GOODS_ISSUE(3) → SHIP(4)
- All seed routes have `is_system = 1`
- Only one route per org has `is_default = 1` (enforced by trigger or application logic)
- EF Core entities and query filter configured

M2: Product Variant Route Assignment

Table: `inventory.ProductVariants` (ALTER)

```
ALTER TABLE inventory.ProductVariants
ADD fulfillment_route_id INT NULL
CONSTRAINT FK_ProductVariants_FulfillmentRoute
FOREIGN KEY REFERENCES logistics.FulfillmentRoutes(fulfillment_route_id);
```

- Column `fulfillment_route_id` exists as `INT NULL` on `inventory.ProductVariants`
- FK constraint to `logistics.FulfillmentRoutes` exists
- EF Core entity `ProductVariant` has `FulfillmentRouteId` and navigation property

M3: SO Line Route Assignment**Table: `demand.SaleOrderLines` (ALTER)**

```
ALTER TABLE demand.SaleOrderLines
ADD fulfillment_route_id INT NULL
CONSTRAINT FK_SaleOrderLines_FulfillmentRoute
FOREIGN KEY REFERENCES logistics.FulfillmentRoutes(fulfillment_route_id);
```

- Column `fulfillment_route_id` exists as `INT NULL` on `demand.SaleOrderLines`
- FK constraint to `logistics.FulfillmentRoutes` exists
- EF Core entity `SaleOrderLine` updated

M4: Delivery Order ↔ Sale Order & Route Link**Tables: `logistics.delivery_orders`, `logistics.delivery_order_lines` (ALTER)**

```
ALTER TABLE logistics.delivery_orders
ADD sale_order_id BIGINT NULL
CONSTRAINT FK_DeliveryOrders_SaleOrder
FOREIGN KEY REFERENCES demand.SaleOrders(sale_order_id);

ALTER TABLE logistics.delivery_orders
ADD fulfillment_route_id INT NULL
CONSTRAINT FK_DeliveryOrders_FulfillmentRoute
FOREIGN KEY REFERENCES logistics.FulfillmentRoutes(fulfillment_route_id);

ALTER TABLE logistics.delivery_order_lines
ADD sale_order_line_id BIGINT NULL
CONSTRAINT FK_DOLines_SaleOrderLine
FOREIGN KEY REFERENCES demand.SaleOrderLines(sale_order_line_id);

-- Extend from_source_type CHECK to include SALE_ORDER
-- (Implementation depends on existing constraint definition)
```

- Column `sale_order_id` exists as `BIGINT NULL` on `logistics.delivery_orders` with FK
- Column `fulfillment_route_id` exists as `INT NULL` on `logistics.delivery_orders` with FK
- Column `sale_order_line_id` exists as `BIGINT NULL` on `logistics.delivery_order_lines` with FK
- `from_source_type` CHECK constraint updated to include `'SALE_ORDER'`
- Indexes created on FK columns

1.4 Addendum 34 Migrations (M1–M4)**M1: Route Category Column & Manufacturing Seed Routes****Table: `logistics.FulfillmentRoutes` (ALTER)**

```
ALTER TABLE logistics.FulfillmentRoutes
ADD route_category NVARCHAR(20) NOT NULL DEFAULT 'STOCK'
CONSTRAINT CK_FulfillmentRoutes_Category
CHECK (route_category IN ('STOCK', 'MANUFACTURE', 'BUY', 'DROPSHIP'));

-- Update existing seed routes to STOCK
UPDATE logistics.FulfillmentRoutes
SET route_category = 'STOCK'
WHERE code IN ('PICK_ONLY', 'PICK_AND_SHIP', 'PICK_PACK_SHIP');
```

```

-- Insert manufacturing routes for all existing orgs (cursor-based)
DECLARE @org_id INT;
DECLARE org_cursor CURSOR FOR SELECT org_id FROM auth.Organizations;
OPEN org_cursor;
FETCH NEXT FROM org_cursor INTO @org_id;
WHILE @@FETCH_STATUS = 0
BEGIN
    -- MFG_PICK_SHIP
    INSERT INTO logistics.FulfillmentRoutes (org_id, code, name, route_category,
        is_default, is_active, is_system, requires_packing, requires_shipping)
    VALUES (@org_id, 'MFG_PICK_SHIP', 'Manufacture, Pick and Ship', 'MANUFACTURE',
        0, 1, 1, 0, 1);
    -- steps: PICK(1), GOODS_ISSUE(2), SHIP(3)

    -- MFG_PICK_PACK_SHIP
    INSERT INTO logistics.FulfillmentRoutes (org_id, code, name, route_category,
        is_default, is_active, is_system, requires_packing, requires_shipping)
    VALUES (@org_id, 'MFG_PICK_PACK_SHIP', 'Manufacture, Pick, Pack and Ship', 'MANUFACTURE',
        0, 1, 1, 1, 1);
    -- steps: PICK(1), PACK(2), GOODS_ISSUE(3), SHIP(4)

    FETCH NEXT FROM org_cursor INTO @org_id;
END;
CLOSE org_cursor; DEALLOCATE org_cursor;

```

All Routes After Migration:

Code	Category	Default	Steps
PICK_ONLY	STOCK	No	Pick → GI
PICK_AND_SHIP	STOCK	Yes	Pick → GI → Ship
PICK_PACK_SHIP	STOCK	No	Pick → Pack → GI → Ship
MFG_PICK_SHIP	MANUFACTURE	No	Pick → GI → Ship
MFG_PICK_PACK_SHIP	MANUFACTURE	No	Pick → Pack → GI → Ship

- Column `route_category` exists as `NVARCHAR(20) NOT NULL DEFAULT 'STOCK'`
- CHECK constraint `CK_FulfillmentRoutes_Category` allows only `STOCK/MANUFACTURE/BUY/DROPSHIP`
- Existing routes updated to `route_category = 'STOCK'`
- Route `MFG_PICK_SHIP` inserted for all existing orgs with `route_category = 'MANUFACTURE'`
- Route `MFG_PICK_PACK_SHIP` inserted for all existing orgs with `route_category = 'MANUFACTURE'`
- Steps correctly inserted for both manufacturing routes
- New org creation logic also seeds manufacturing routes
- EF Core entity `FulfillmentRoute` has `RouteCategory` property

M2: Lead Time Defaults & Variant Lead Time Columns

Tables: `lookups.LeadTimeDefaults` (CREATE), `inventory.ProductVariants` (ALTER)

```

CREATE TABLE lookups.LeadTimeDefaults (
    lead_time_default_id INT IDENTITY(1,1) PRIMARY KEY,
    org_id INT NOT NULL UNIQUE REFERENCES auth.Organizations(org_id),
    pick_pack_days INT NOT NULL DEFAULT 1,
    shipping_lead_time_days INT NOT NULL DEFAULT 3,
    sales_buffer_days INT NOT NULL DEFAULT 1,
    manufacturing_buffer_days INT NOT NULL DEFAULT 0,
    quality_inspection_days INT NOT NULL DEFAULT 0,
    internal_transfer_days INT NOT NULL DEFAULT 0,
    created_by INT NOT NULL,
    created_at DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    modified_by INT NULL,
    modified_at DATETIME2 NULL
);

ALTER TABLE inventory.ProductVariants ADD supplier_lead_time_days INT NULL;
ALTER TABLE inventory.ProductVariants ADD manufacturing_lead_time_days INT NULL;
ALTER TABLE inventory.ProductVariants ADD manufacturing_buffer_days INT NULL;

```

```

ALTER TABLE inventory.ProductVariants ADD quality_inspection_days INT NULL;
ALTER TABLE inventory.ProductVariants ADD internal_transfer_days INT NULL;
ALTER TABLE inventory.ProductVariants ADD pick_pack_days INT NULL;
ALTER TABLE inventory.ProductVariants ADD shipping_lead_time_days INT NULL;
ALTER TABLE inventory.ProductVariants ADD sales_buffer_days INT NULL;

-- Backfill from existing lead_time_days (if column existed)
UPDATE inventory.ProductVariants
  SET supplier_lead_time_days = lead_time_days
  WHERE lead_time_days IS NOT NULL;

```

- Table `lookups.LeadTimeDefaults` created with all columns
- Unique constraint on `org_id` (one row per org)
- Default values: `pick_pack=1`, `shipping=3`, `sales_buffer=1`, `mfg_buffer=0`, `quality=0`, `transfer=0`
- Seed row inserted for each existing org
- 8 new columns on `inventory.ProductVariants`:
- `supplier_lead_time_days` INT NULL
- `manufacturing_lead_time_days` INT NULL
- `manufacturing_buffer_days` INT NULL
- `quality_inspection_days` INT NULL
- `internal_transfer_days` INT NULL
- `pick_pack_days` INT NULL
- `shipping_lead_time_days` INT NULL
- `sales_buffer_days` INT NULL
- Existing `lead_time_days` values migrated to `supplier_lead_time_days`
- EF Core entities updated

M3: Lead Time Columns on Demand Line Tables

Tables: `demand.SaleInquiryLines`, `demand.SaleQuotationLines`, `demand.SaleOrderLines` (ALTER)

```

-- Applied to each of the three tables:
ALTER TABLE demand.{TableName} ADD calculated_lead_time_days INT NULL;
ALTER TABLE demand.{TableName} ADD calculated_delivery_date DATE NULL;
ALTER TABLE demand.{TableName} ADD manual_delivery_date DATE NULL;
ALTER TABLE demand.{TableName} ADD lead_time_calculated_at DATETIME2 NULL;
ALTER TABLE demand.{TableName} ADD effective_delivery_date
  AS COALESCE(manual_delivery_date, calculated_delivery_date) PERSISTED;

```

- On `demand.SaleInquiryLines` (referred to as InquiryLines):
- `calculated_lead_time_days` INT NULL
- `calculated_delivery_date` DATE NULL
- `manual_delivery_date` DATE NULL
- `lead_time_calculated_at` DATETIME2 NULL
- `effective_delivery_date` computed column (PERSISTED)
- On `demand.SaleQuotationLines` (referred to as QuotationLines):
- `calculated_lead_time_days` INT NULL
- `calculated_delivery_date` DATE NULL
- `manual_delivery_date` DATE NULL
- `lead_time_calculated_at` DATETIME2 NULL
- `effective_delivery_date` computed column (PERSISTED)
- On `demand.SaleOrderLines`:
- `calculated_lead_time_days` INT NULL
- `calculated_delivery_date` DATE NULL
- `manual_delivery_date` DATE NULL
- `lead_time_calculated_at` DATETIME2 NULL
- `effective_delivery_date` computed column (PERSISTED)
- COALESCE logic: `manual_delivery_date` takes priority over `calculated_delivery_date`
- EF Core entities updated (computed column mapped as `ValueGeneratedOnAddOrUpdate`)

M4: Production Order Route & Delivery Link

Table: `material.ProductionOrders` (ALTER)

```
ALTER TABLE material.ProductionOrders
  ADD fulfillment_route_id INT NULL
  CONSTRAINT FK_ProductionOrders_FulfillmentRoute
  FOREIGN KEY REFERENCES logistics.FulfillmentRoutes(fulfillment_route_id);

ALTER TABLE material.ProductionOrders
  ADD delivery_order_id BIGINT NULL
  CONSTRAINT FK_ProductionOrders_DeliveryOrder
  FOREIGN KEY REFERENCES logistics.delivery_orders(delivery_order_id);

CREATE INDEX IX_ProductionOrders_FulfillmentRoute
  ON material.ProductionOrders(fulfillment_route_id)
  WHERE fulfillment_route_id IS NOT NULL;

CREATE INDEX IX_ProductionOrders_DeliveryOrder
  ON material.ProductionOrders(delivery_order_id)
  WHERE delivery_order_id IS NOT NULL;

-- Backfill existing POs linked to MANUFACTURE routes
UPDATE po
  SET po.fulfillment_route_id = sol.fulfillment_route_id
  FROM material.ProductionOrders po
  JOIN demand.SaleOrderLines sol ON po.sale_order_line_id = sol.sale_order_line_id
  WHERE po.sale_order_line_id IS NOT NULL
  AND sol.fulfillment_route_id IS NOT NULL;
```

- Column `fulfillment_route_id` exists as `INT NULL` on `material.ProductionOrders` with FK
- Column `delivery_order_id` exists as `BIGINT NULL` on `material.ProductionOrders` with FK
- Filtered index on `fulfillment_route_id` exists
- Filtered index on `delivery_order_id` exists
- Backfill applied for existing POs linked to SO lines with routes
- EF Core entity `ProductionOrder` has `FulfillmentRouteId` and `DeliveryOrderId` properties

Permission Claims — Addendum 34 (M4 continued)

```
INSERT INTO auth.Permissions (permission_name, description, module)
VALUES ('lead_time_defaults_manage', 'Manage organization lead time defaults', 'lookups');
```

- Permission `lead_time_defaults_manage` inserted
- Available in role assignment

Permission Claims — Addendum 33

```
INSERT INTO auth.Permissions (permission_name, description, module)
VALUES
  ('fulfillment_route_view', 'View fulfillment routes', 'logistics'),
  ('fulfillment_route_manage', 'Manage fulfillment route configuration', 'logistics'),
  ('fulfillment_route_assign', 'Assign routes to products and orders', 'logistics');
```

- Permission `fulfillment_route_view` inserted
- Permission `fulfillment_route_manage` inserted
- Permission `fulfillment_route_assign` inserted
- Applied to correct controllers

Consolidated Permission Claims (14 Total)

#	Claim	Module	Addendum
1	sales_inquiry_view	demand	32
2	sales_inquiry_create	demand	32
3	sales_inquiry_edit	demand	32
4	sales_quotation_view	demand	32
5	sales_quotation_create	demand	32
6	sales_quotation_edit	demand	32
7	sales_quotation_send	demand	32
8	sales_order_create	demand	32
9	sales_reserve_inventory	demand	32
10	sales_release_reservation	demand	32
11	fulfillment_route_view	logistics	33
12	fulfillment_route_manage	logistics	33
13	fulfillment_route_assign	logistics	33
14	lead_time_defaults_manage	lookups	34

- All 14 permission claims exist in `auth.Permissions`
- Each is assignable to roles via the admin UI
- Each is enforced by `[Authorize]` on the correct API endpoints

PART 2 — BACKEND SERVICES & BUSINESS LOGIC

2.1 Addendum 31 — Manufacturing Flow Corrections (10 Changes)

C1: Sale Order Min/Max Quantity Limits

Service: `SaleOrderLineService` (or equivalent validation layer)

Business Rules:

Rule	Description	Verified
BR-C1-01	If <code>sale_order_min_qty</code> is set on product AND ordered qty < min → block with message showing the minimum	[]
BR-C1-02	If <code>sale_order_max_qty</code> is set on product AND ordered qty > max → block with message showing the maximum	[]
BR-C1-03	Min/Max validation runs at SO line add/edit, not at confirmation	[]
BR-C1-04	NULL min/max means no limit enforced	[]
BR-C1-05	On Product save: if both set, min must be ≤ max	[]
BR-C1-06	Min/Max displayed as helper text on SO line form	[]

- Validation implemented in correct service
- Error messages include the actual min/max values
- Cross-validation on Product save prevents min > max

C2: Product Tab Consolidation

UI Changes Only — No schema change

- Manufacturing tab removed from Product form
- Classification tab now has two sections:
 - "Product Classification" (existing fields)
 - "Manufacturing & Supply" (moved from Manufacturing tab)
- All existing Classification fields retained
- All existing Manufacturing fields retained in new location
- No data migration needed (same underlying fields)

C3: Draft PO Flow from Supply Requirement Engine

Service: `SupplyRequirementEngine`

Business Rules:

Rule	Description	Verified
BR-C3-01	Engine creates PO in DRAFT status (not auto-submitted)	[]

Rule	Description	Verified
BR-C3-02	Default supplier selection priority: product's default supplier → last supplier used for this product → NULL (no supplier, user must select)	[]
BR-C3-03	If no default supplier found, PO created with supplier = NULL, flagged for user attention	[]
BR-C3-04	Notification sent to users with <code>purchase_order_write</code> permission	[]
BR-C3-05	Draft PO includes source reference (Production Order that triggered it)	[]
BR-C3-06	Duplicate prevention: do not create Draft PO if one already exists for same product + Production Order	[]
BR-C3-07	Quantities on Draft PO match shortage quantities from material availability check	[]

- `SupplyRequirementEngine` creates POs with `Status = DRAFT`
- Supplier resolution follows the 3-tier priority
- MediatR notification event published for purchase team
- Duplicate check queries existing Draft POs before creation

C4: Inline BOM on Product Form

UI Changes Only — No schema change

- Standalone BOM list page removed (or hidden)
- Standalone BOM edit page removed (or hidden)
- Product form shows BOM section with two-panel layout:
 - Left panel (35%): BOM list for this product
 - Right panel (65%): BOM editor (selected BOM details)
- BOM mandatory for manufacturing products (BR-C4-01: cannot save manufacturing product without at least one BOM)
- BOM section visibility: only for products classified as manufacturing (BR-C4-02)
- Workflow buttons shown by BOM status (Draft → Active → Obsolete)
- Create New BOM button in left panel
- Delete BOM only if in Draft status

C5: BOM Effective Date Auto-Default

Application-level change only — No schema change

- `effective_from` defaults to today's date when creating new BOM (BR-C5-01)
- Default set at application level (DTO mapping or form initialization), not DB default
- User can override the default
- Existing BOMs unaffected

C6: Remove Warehouse from BOM

Service/Entity Changes:

- `BillOfMaterialLine` entity no longer has `WarehouseId` property (BR-C6-01)
- PMR (Production Material Requirement) warehouse always sourced from `ProductionOrder.production_warehouse_id`
- All code that previously read `BOMLine.warehouse_id` updated
- BOM line form no longer shows warehouse dropdown
- Migration M3 executed (DROP COLUMN)

C7: Production Order ↔ Sale Order Link

Service: Production Order creation logic

- When PO created from SO confirmation, `sale_order_id` and `sale_order_line_id` populated (BR-C7-01)
- Navigation from PO detail → linked SO is functional
- Navigation from SO detail → linked PO(s) is functional
- PO list can filter by "Has Sale Order" / "Standalone"

C8: Material Availability Tab

Service: `IMaterialReadinessService`

Business Rules:

Rule	Description	Verified
BR-C8-01	Reserve action calls <code>IStockReservationService.Reserve()</code> for available materials	[]
BR-C8-02	Allocation Request creates a request to source material from other warehouses or via purchase	[]

UI:

- Material Availability tab on Production Order detail page
- "Available" section lists materials with stock, shows Reserve button
- "Shortage" section lists materials without sufficient stock, shows Allocation Request button
- "Reserve All Available" batch action button
- Allocation Request dialog with pre-filled fields (product, qty needed, preferred warehouse)
- GET `/api/production-orders/{id}/material-availability` returns availability data
- POST `/api/production-orders/{id}/material-availability/reserve` executes reservation

C9: Consolidated Purchase Required Dashboard

Service: Purchase Required aggregation query

Business Rules:

Rule	Description	Verified
BR-C9-01	Aggregates shortages across ALL active Production Orders per product per org	[]
BR-C9-02	Duplicate PO prevention: if Draft PO already exists for product, show "PO Pending" badge instead of "Create PO" button	[]

- GET `/api/purchase-required` endpoint implemented
- Dashboard at Manufacturing → Purchase Required menu
- Groups by product, sums shortage quantities across POs
- Shows affected PO count per product
- Clicking a product row opens detail drawer showing affected POs
- "Create Draft PO" action checks for existing Draft POs (duplicate prevention)

C10: GRN → Allocation → Reservation Flow

Services: `IAllocationEngine`, `FGRConfirmedEventHandler`

Business Rules:

Rule	Description	Verified
BR-C10-01	GRN receipt → allocation is MANUAL (user triggers via POST <code>/api/allocation/run</code>)	[]
BR-C10-02	FGR (Finished Goods Receipt) → allocation is AUTOMATIC (MediatR event handler)	[]
BR-C10-03	After allocation, material status on PO updated (from SHORTAGE to AVAILABLE or PARTIAL)	[]
BR-C10-04	Full allocation of all materials transitions PO material status to READY	[]

- POST `/api/allocation/run` endpoint exists for manual allocation
- `AllocationRunCommand` MediatR handler implemented
- `FGRConfirmedEvent` event handler performs automatic allocation
- Material status updates correctly after allocation
- Partial allocation shows as PARTIAL, full as READY

2.2 Addendum 32 — Sales Pre-Order Pipeline (5 Changes)

C1: Sale Inquiry

Service: `SaleInquiryService`

State Machine: RECEIVED → UNDER_REVIEW → REVIEW_COMPLETE → QUOTED → DECLINED

Business Rules:

Rule	Description	Verified
BR-C1-01	Inquiry number auto-generated: <code>INQ-{YYYY}-{NNNN}</code> (sequential per org per year)	[]
BR-C1-02	New inquiry starts in RECEIVED status	[]
BR-C1-03	Transition to UNDER_REVIEW requires <code>assigned_to</code> to be set	[]
BR-C1-04	Line evaluation status starts as PENDING	[]
BR-C1-05	REVIEW_COMPLETE requires all lines to have non-PENDING evaluation	[]
BR-C1-06	QUOTED transition creates a linked Quotation (auto or manual)	[]
BR-C1-07	DECLINED requires at least one line with <code>CANNOT_SUPPLY</code> or rejection reason	[]

API Endpoints:

- GET `/api/sale-inquiries` — list with filters (status, customer, date range)
- GET `/api/sale-inquiries/{id}` — detail with lines
- POST `/api/sale-inquiries` — create
- PUT `/api/sale-inquiries/{id}` — update
- PUT `/api/sale-inquiries/{id}/status` — transition status
- PUT `/api/sale-inquiries/{id}/lines/{lineId}/evaluate` — set line evaluation

C2: Sale Quotation

Service: `SaleQuotationService`

State Machine: DRAFT → SENT → ACCEPTED → REJECTED → EXPIRED → CONVERTED

Business Rules:

Rule	Description	Verified
BR-C2-01	Quotation number auto-generated: <code>SQ-{YYYY}-{NNNN}</code> (sequential per org per year)	[]
BR-C2-02	New quotation starts in DRAFT status	[]
BR-C2-03	SENT transition requires at least one NORMAL line with price > 0	[]
BR-C2-04	SENT transition records <code>sent_at</code> timestamp	[]
BR-C2-05	Customer response per line: PENDING / ACCEPTED / REJECTED / COUNTER	[]
BR-C2-06	COUNTER response requires <code>counter_price</code> or <code>counter_qty</code> to be set	[]
BR-C2-07	ACCEPTED status requires at least one line with <code>customer_response = ACCEPTED</code>	[]
BR-C2-08	ALTERNATIVE lines link to parent via <code>parent_line_id</code>	[]
BR-C2-09	REJECTED lines require <code>rejection_reason_id</code>	[]
BR-C2-10	EXPIRED set automatically by <code>QuotationExpiryJob</code> when <code>valid_until < today</code>	[]
BR-C2-11	CONVERTED transition creates linked SO with only ACCEPTED lines	[]

Hangfire Job:

- `QuotationExpiryJob` registered in Hangfire
- Runs daily at 01:00 UTC
- Updates SENT quotations where `valid_until < GETUTCDATE()` to EXPIRED
- Does not expire DRAFT quotations

Quotation → SO Conversion (BR-C2-11):

- Only lines where `customer_response = ACCEPTED` flow to SO
- Lines with `customer_response = COUNTER` require seller acceptance of counter-price first
- SO `source_quotation_id` populated
- SO `source_type = 'FROM_QUOTATION'`
- Quotation status transitions to CONVERTED after SO creation
- Original quotation becomes read-only after CONVERTED

API Endpoints:

- GET `/api/sale-quotations` — list
- GET `/api/sale-quotations/{id}` — detail
- POST `/api/sale-quotations` — create
- PUT `/api/sale-quotations/{id}` — update
- PUT `/api/sale-quotations/{id}/send` — transition to SENT
- PUT `/api/sale-quotations/{id}/lines/{lineId}/response` — record customer response
- POST `/api/sale-quotations/{id}/convert` — convert to SO

C3: Sale Order Source Linking & Customer PO

Service: `SaleOrderService` modifications

Business Rules:

Rule	Description	Verified
BR-C3-01	<code>source_type</code> auto-set based on creation path (MANUAL, FROM_QUOTATION, PORTAL, INTER_TENANT)	[]
BR-C3-02	<code>source_quotation_id</code> populated when created from quotation conversion	[]
BR-C3-03	<code>source_inquiry_id</code> populated when traceable back to inquiry	[]
BR-C3-04	<code>customer_po_reference</code> is free-text, max 50 chars	[]
BR-C3-05	<code>customer_po_attachment_id</code> links to <code>SaleOrderAttachments</code> record	[]

- SO creation from quotation sets `source_type = 'FROM_QUOTATION'`
- SO detail page shows source document link (clickable)
- Customer PO section on SO form: reference field + date picker + file upload
- Source type displayed but not user-editable
- Breadcrumb navigation: Inquiry → Quotation → Sale Order

C4: Delivery Indicators & RBAC Reservation

Service: Delivery indicator computation (read-time, not persisted)

Indicator Logic (computed at read time):

Color	Condition
GREEN	Line fully fulfilled (<code>delivered_qty</code> >= <code>ordered_qty</code>)
BLUE	Fully reserved (<code>reserved_qty</code> >= <code>ordered_qty</code> , not yet fulfilled)
YELLOW	Partially reserved (<code>0</code> < <code>reserved_qty</code> < <code>ordered_qty</code>)
RED	Nothing reserved and nothing fulfilled (<code>reserved_qty</code> = <code>0</code> AND <code>delivered_qty</code> = <code>0</code>)
GREY	Line cancelled

Business Rules:

Rule	Description	Verified
BR-C4-01	Delivery indicator computed at read time (DTO mapping), NOT persisted	[]
BR-C4-02	Reserve action requires <code>sales_reserve_inventory</code> permission	[]
BR-C4-03	Release action requires <code>sales_release_reservation</code> permission	[]
BR-C4-04	Reserve calls <code>IStockReservationService.Reserve()</code> with <code>source = SALES_ORDER</code>	[]
BR-C4-05	Release calls <code>IStockReservationService.Release()</code>	[]
BR-C4-06	<code>reserved_qty</code> updated on <code>SaleOrderLines</code> after reserve/release	[]
BR-C4-07	SO cancellation auto-releases all reservations on that SO's lines	[]
BR-C4-08	Reserve qty cannot exceed (<code>ordered_qty</code> - <code>delivered_qty</code>)	[]

- Indicator displayed on SO line list (colored dot/badge)
- Reserve button per line (or "Reserve All" bulk action)
- Release button per line

- Permissions enforced via `[Authorize]`
- Auto-release on SO cancellation implemented

C5: Rejection Reason Lookup

Business Rules:

Rule	Description	Verified
BR-C5-01	System codes (<code>is_system=1</code>) cannot be deleted	[]
BR-C5-02	Users can add custom rejection reasons (<code>is_system=0</code>)	[]
BR-C5-03	Rejection reasons shared between Inquiry and Quotation lines	[]

- CRUD API for rejection reasons exists
- System codes protected from deletion
- Dropdown populated in both Inquiry line and Quotation line forms
- Lookup management page accessible from Settings/Lookups

2.3 Addendum 33 — Fulfillment Routes (5 Changes)

C1: Fulfillment Route Configuration

Business Rules:

Rule	Description	Verified
BR-C1-01	Step codes constrained to: PICK, PACK, STAGE, APPROVAL, GOODS_ISSUE, SHIP	[]
BR-C1-02	Steps ordered by <code>step_order</code> (sequential, no gaps required)	[]
BR-C1-03	Exactly one route per org can be <code>is_default = 1</code>	[]
BR-C1-04	System routes (<code>is_system = 1</code>) cannot be deleted	[]
BR-C1-05	System routes can be deactivated (<code>is_active = 0</code>) but not deleted	[]
BR-C1-06	Route code unique per org	[]
BR-C1-07	At least one step required per route	[]
BR-C1-08	PICK step should be present in all routes (warning if missing, not enforced)	[]
BR-C1-09	Custom routes: user can create with any combination of allowed steps	[]

API Endpoints:

- GET `/api/fulfillment-routes` — list all routes for org
- GET `/api/fulfillment-routes/{id}` — detail with steps
- POST `/api/fulfillment-routes` — create custom route
- PUT `/api/fulfillment-routes/{id}` — update route
- DELETE `/api/fulfillment-routes/{id}` — delete (only non-system)
- PUT `/api/fulfillment-routes/{id}/set-default` — mark as org default

UI:

- Route configuration page at Settings → Fulfillment Routes

- Route list with columns: Code, Name, Steps (visual), Default badge, Status
- Route editor: name, code, description, step builder (drag-and-drop or ordered list)
- Step builder shows available step codes as chips to add/remove/reorder
- System routes show lock icon, edit limited to activate/deactivate

C2: Product Variant Route Assignment

Business Rules:

Rule	Description	Verified
BR-C2-01	Variant <code>fulfillment_route_id</code> is optional (NULL means "use org default")	[]
BR-C2-02	Bulk assignment API: assign route to all variants in a product category	[]

- Fulfillment Route dropdown on Product Variant form
- Shows active routes only
- NULL option labeled "Use Organization Default"
- Bulk assignment endpoint: POST `/api/product-variants/bulk-assign-route`

C3: SO Line Route Selection & Confirmation Gate

Service: `EffectiveRouteResolver`

4-Level Route Resolution Priority:

1. SO line explicit override (`SaleOrderLines.fulfillment_route_id`)
2. Product variant default (`ProductVariants.fulfillment_route_id`)
3. Organization default route (`FulfillmentRoutes.is_default = 1`)
4. NULL → **blocks SO confirmation**

Business Rules:

Rule	Description	Verified
BR-C3-01	Route resolution follows the 4-level priority	[]
BR-C3-02	SO confirmation BLOCKED if any line resolves to NULL route	[]
BR-C3-03	Confirmation error message identifies which lines have no route	[]
BR-C3-04	Route dropdown on SO line form shows active routes	[]
BR-C3-05	Delivery Preview (<code>SaleOrderDeliveryPreviewService</code>) computed on-the-fly, not persisted	[]

- `EffectiveRouteResolver` class/service implemented
- Resolution follows exact 4-level priority
- `SaleOrderService.ConfirmAsync()` calls resolver for each line
- Confirmation fails with clear error if any line has no route
- Delivery Preview endpoint: GET `/api/sale-orders/{id}/delivery-preview`
- Preview groups lines by effective route and shows expected delivery count

C4: Delivery Order Creation from SO

Service: `SaleOrderDeliveryCreator`

Business Rules:

Rule	Description	Verified
BR-C4-01	SO confirmation creates Delivery Orders grouped by effective fulfillment route	[]
BR-C4-02	One DO per route group (lines with same effective route go into one DO)	[]
BR-C4-03	DO created in DRAFT status	[]
BR-C4-04	DO <code>from_source_type = 'SALE_ORDER'</code>	[]
BR-C4-05	DO <code>sale_order_id</code> and <code>fulfillment_route_id</code> populated	[]
BR-C4-06	DO lines link back via <code>sale_order_line_id</code>	[]
BR-C4-07	SO cancellation auto-cancels linked DOs not yet GOODS_ISSUED; post-GOODS_ISSUED DOs require manual reversal	[]

- `SaleOrderDeliveryCreator` service implemented
- Grouping logic: lines with same effective route → one DO
- DO created in DRAFT status
- All FK links populated (`sale_order_id`, `fulfillment_route_id`, `sale_order_line_id`)
- SO cancellation handler checks DO status before auto-cancelling
- Error message when trying to cancel SO with post-GOODS_ISSUED DOs

C5: Route-Aware State Machine

Service: `RouteAwareDeliveryStateMachine`

Step-to-Status Mapping:

Step Code	Statuses Covered
PICK	RELEASED → PICKING → PICKED
PACK	PACKED
STAGE	STAGED
APPROVAL	PENDING_APPROVAL
GOODS_ISSUE	GOODS_ISSUED
SHIP	IN_TRANSIT → OUT_FOR_DELIVERY → DELIVERED

Business Rules:

Rule	Description	Verified
BR-C5-01	State machine reads route steps and skips statuses whose <code>step_code</code> is absent	[]
BR-C5-02	Skipping is transparent: transition goes directly from last status of prior step to first status of next step	[]
BR-C5-03	Route with only PICK + GOODS_ISSUE skips PACK, STAGE, APPROVAL, and all SHIP statuses	[]
BR-C5-04	All 15 existing statuses remain in the enum; skipping is per-delivery, not per-system	[]
BR-C5-05	Manual status override remains available for admin users	[]

- `RouteAwareDeliveryStateMachine` class implemented
- Constructor or initialization loads route steps for the delivery's route
- `GetNextStatus()` method skips statuses not in the route
- `GetPreviousStatus()` method also respects route steps

- `CanTransitionTo()` method validates against route-allowed statuses
- Tested with each of the 3 seed routes (PICK_ONLY, PICK_AND_SHIP, PICK_PACK_SHIP)
- Tested with 2 manufacturing routes (MFG_PICK_SHIP, MFG_PICK_PACK_SHIP) from Addendum 34

2.4 Addendum 34 — Route Classification & Production-Delivery Bridge (6 Changes)

C1: Route Category

Business Rules:

Rule	Description	Verified
BR-C1-01	<code>route_category</code> must be one of: STOCK, MANUFACTURE, BUY, DROPSHIP	[]
BR-C1-02	Existing PICK_ONLY, PICK_AND_SHIP, PICK_PACK_SHIP routes set to STOCK	[]
BR-C1-03	New MFG_PICK_SHIP and MFG_PICK_PACK_SHIP routes set to MANUFACTURE	[]
BR-C1-04	Route category displayed on route configuration UI and is user-selectable for custom routes	[]

- Category dropdown on route create/edit form
- Category badge visible in route list
- Seed route categories correctly set
- New org creation seeds both STOCK and MANUFACTURE routes

C2: Route-Based Product Classification

Business Rules:

Rule	Description	Verified
BR-C2-01	BOM tab visible when product variant's route has <code>route_category = 'MANUFACTURE'</code>	[]
BR-C2-02	<code>is_manufacturable</code> flag auto-synced from route category (TRUE if MANUFACTURE, FALSE otherwise)	[]
BR-C2-03	Legacy boolean filters (<code>is_manufacturable</code>) still work for backward compatibility	[]
BR-C2-04	Changing variant route from MANUFACTURE to STOCK hides BOM tab (but does not delete existing BOMs)	[]

- BOM tab visibility logic updated from boolean check to route category check
- `is_manufacturable` set automatically when route assigned/changed
- Existing API filters using `is_manufacturable` continue to work
- BOMs preserved when route changes away from MANUFACTURE (hidden but not deleted)

C3: Lead Time Management

Service: Lead time resolution logic

8 Lead Time Components:

Component	Column	Route Types	Default
Supplier Lead Time	<code>supplier_lead_time_days</code>	BUY, STOCK (purchased items)	From variant
Manufacturing Lead Time	<code>manufacturing_lead_time_days</code>	MANUFACTURE	From variant
Manufacturing Buffer	<code>manufacturing_buffer_days</code>	MANUFACTURE	0
Quality Inspection	<code>quality_inspection_days</code>	All	0
Internal Transfer	<code>internal_transfer_days</code>	All (multi-warehouse)	0
Pick & Pack	<code>pick_pack_days</code>	All	1
Shipping Lead Time	<code>shipping_lead_time_days</code>	All (with shipping step)	3
Sales Buffer	<code>sales_buffer_days</code>	All	1

Business Rules:

Rule	Description	Verified
BR-C3-01	Variant-level lead times override org defaults (NULL means "use org default")	[]
BR-C3-02	Org defaults managed via <code>lookups.LeadTimeDefaults</code> (one row per org)	[]
BR-C3-03	Lead time component visibility on variant form driven by route type	[]
BR-C3-04	MANUFACTURE route shows: <code>manufacturing_lead_time_days</code> , <code>manufacturing_buffer_days</code> , <code>quality_inspection_days</code> , <code>pick_pack_days</code> , <code>shipping</code> , <code>sales_buffer</code>	[]
BR-C3-05	STOCK route shows: <code>supplier_lead_time_days</code> , <code>quality_inspection_days</code> , <code>pick_pack_days</code> , <code>shipping</code> , <code>sales_buffer</code>	[]

- Lead Time Defaults management page at Settings → Lead Time Defaults
- Requires `lead_time_defaults_manage` permission
- Variant form shows lead time fields appropriate to route type
- Resolution: variant override → org default → hardcoded fallback
- Org defaults seeded for existing orgs

C4: Lead Time Calculation (BOM-Aware)**Service:** `ILeadTimeCalculator`**Algorithm:**

```
CalculateLeadTime(variant_id, quantity):
  1. Check stock: if variant has sufficient stock → lead = 0, return
  2. Get route category for variant
```

3. If MANUFACTURE:
 - a. Get BOM for variant
 - b. For each sub-component:
 - Recursively calculate lead time (parallel → take MAX)
 - c. Get this-level operations:
 - Sum: manufacturing_lead_time + mfg_buffer + quality + internal_transfer
 - d. Convert BOM operation times: 480 min = 1 working day
 - e. Total = MAX(sub-component leads) + this-level sum + pick_pack + shipping + sales_buffer
4. If STOCK/BUY:
 - Total = supplier_lead_time + quality + internal_transfer + pick_pack + shipping + sales_buffer
5. Cache result for 5 minutes per (variant_id, quantity_bucket)

Business Rules:

Rule	Description	Verified
BR-C4-01	Recursive BOM traversal: sub-components run in parallel (take MAX of their lead times)	[]
BR-C4-02	This-level operations are sequential (SUM)	[]
BR-C4-03	Stock-aware: if component has sufficient stock, its lead time = 0	[]
BR-C4-04	Cycle detection via visited HashSet — cycle returns 0 and logs warning	[]
BR-C4-05	5-minute cache per (variant_id , quantity_bucket) using IMemoryCache	[]
BR-C4-06	BOM operation times converted at 480 minutes per working day	[]

- **ILeadTimeCalculator** interface defined
- **LeadTimeCalculator** implementation with recursive BOM traversal
- Cycle detection prevents infinite recursion
- Cache layer with 5-minute TTL
- Stock check integrated (calls inventory service)
- Calculation triggered when:
 - Inquiry line added/changed → **calculated_lead_time_days** and **calculated_delivery_date** updated
 - Quotation line added/changed → same columns updated
 - SO line added/changed → same columns updated
- **calculated_delivery_date = today + calculated_lead_time_days**
- **effective_delivery_date** computed column works correctly (manual overrides calculated)

C5: SO Confirmation Split by Route Category

Service: `SaleOrderService.ConfirmAsync()` (modified)

Split Logic:

- ```
ConfirmAsync(saleOrderId):
```
1. Resolve effective route for each line (EffectiveRouteResolver)
  2. Validate: all lines must have a route (else block — from Addendum 33)
  3. Group lines by route\_category:
    - STOCK lines → SaleOrderDeliveryCreator (existing from Addendum 33)
    - MANUFACTURE lines → SaleOrderProductionCreator (NEW)
  4. Mixed orders: process both groups

**Business Rules:**

| Rule     | Description                                                          | Verified |
|----------|----------------------------------------------------------------------|----------|
| BR-C5-01 | STOCK lines → Delivery Order creation (existing flow)                | [ ]      |
| BR-C5-02 | MANUFACTURE lines → Production Order creation (new flow)             | [ ]      |
| BR-C5-03 | Mixed orders process both — some lines create DOs, others create POs | [ ]      |

| Rule     | Description                                                                                                   | Verified |
|----------|---------------------------------------------------------------------------------------------------------------|----------|
| BR-C5-04 | <code>SaleOrderProductionCreator</code> creates one PO per manufacturing line                                 | [ ]      |
| BR-C5-05 | PO includes: <code>sale_order_id</code> , <code>sale_order_line_id</code> , <code>fulfillment_route_id</code> | [ ]      |
| BR-C5-06 | BOM explosion triggered on PO creation                                                                        | [ ]      |

**SaleOrderProductionCreator Details:**

- Service class `SaleOrderProductionCreator` implemented
- Creates `ProductionOrder` with:
  - `sale_order_id` = SO id
  - `sale_order_line_id` = SO line id
  - `fulfillment_route_id` = line's effective route id
  - `planned_qty` = SO line quantity
  - `product_variant_id` = SO line product
- BOM explosion: creates PMR lines from active BOM for the product
- Backward scheduling: `planned_start_date` = `effective_delivery_date` - `calculated_lead_time_days`
- PO created in DRAFT (or PLANNED) status
- If no active BOM found for a MANUFACTURE product → error, confirmation fails for that line

**C6: Production → Delivery Bridge**

**Service:** `FGRConfirmedEventHandler` (extended), `ProductionDeliveryCreator` (new)

**Flow:**

```

Production Order Complete (FGR Confirmed)
↓
FGRConfirmedEventHandler
 ■■■ Existing: Automatic allocation of finished goods
 ■■■ NEW: If PO has sale_order_id AND fulfillment_route_id
 ↓
 ProductionDeliveryCreator
 ■■■ Creates Delivery Order from completed PO
 ■■■ delivery_qty = P0.actual_qty (reflects actual yield)
 ■■■ D0.fulfillment_route_id = P0.fulfillment_route_id
 ■■■ D0.sale_order_id = P0.sale_order_id
 ■■■ P0.delivery_order_id = new D0 id
 ■■■ Handles shortfall (actual & planned)

```

**Business Rules:**

| Rule     | Description                                                                                                           | Verified |
|----------|-----------------------------------------------------------------------------------------------------------------------|----------|
| BR-C6-01 | FGR confirmed triggers automatic allocation (existing behavior preserved)                                             | [ ]      |
| BR-C6-02 | If PO has <code>sale_order_id</code> AND <code>fulfillment_route_id</code> , auto-create Delivery Order               | [ ]      |
| BR-C6-03 | <code>delivery_qty</code> = <code>P0.actual_qty</code> (actual yield, not planned qty)                                | [ ]      |
| BR-C6-04 | Shortfall: if <code>actual_qty</code> < <code>planned_qty</code> , create delivery for actual qty + send notification | [ ]      |
| BR-C6-05 | Zero production: if <code>actual_qty</code> = 0, no delivery created + error notification sent                        | [ ]      |
| BR-C6-06 | <code>P0.delivery_order_id</code> populated for traceability                                                          | [ ]      |
| BR-C6-07 | DO <code>from_source_type</code> = 'PRODUCTION_ORDER' or extended source type                                         | [ ]      |

| Rule     | Description                                                                                                                                           | Verified |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| BR-C6-08 | MTS escape hatch: salesperson overrides manufacturing route to stock route on SO line → delivery created immediately from stock (bypasses production) | [ ]      |
| BR-C6-09 | Partial production notification includes: PO number, planned vs actual qty, shortfall amount                                                          | [ ]      |

- `FGRConfirmedEventHandler` extended with delivery creation logic
- `ProductionDeliveryCreator` service implemented
- Delivery quantity based on actual production yield
- Shortfall handling: delivery for actual + notification
- Zero production handling: no delivery + error notification
- PO ↔ DO bidirectional link established
- MTS escape hatch: route override on SO line skips production entirely
- Notification service called for shortfall/zero scenarios

## PART 3 — API ENDPOINTS CONSOLIDATED

### 3.1 New Endpoints

| #  | Method | Endpoint                                                  | Addendum | Permission Required      |
|----|--------|-----------------------------------------------------------|----------|--------------------------|
| 1  | GET    | /api/purchase-required                                    | 31       | purchase_order_view      |
| 2  | GET    | /api/production-orders/{id}/material-availability         | 31       | (production order view)  |
| 3  | POST   | /api/production-orders/{id}/material-availability/reserve | 31       | (production order edit)  |
| 4  | POST   | /api/allocation/run                                       | 31       | (inventory manage)       |
| 5  | GET    | /api/sale-inquiries                                       | 32       | sales_inquiry_view       |
| 6  | GET    | /api/sale-inquiries/{id}                                  | 32       | sales_inquiry_view       |
| 7  | POST   | /api/sale-inquiries                                       | 32       | sales_inquiry_create     |
| 8  | PUT    | /api/sale-inquiries/{id}                                  | 32       | sales_inquiry_edit       |
| 9  | PUT    | /api/sale-inquiries/{id}/status                           | 32       | sales_inquiry_edit       |
| 10 | PUT    | /api/sale-inquiries/{id}/lines/{lineId}/evaluate          | 32       | sales_inquiry_edit       |
| 11 | GET    | /api/sale-quotations                                      | 32       | sales_quotation_view     |
| 12 | GET    | /api/sale-quotations/{id}                                 | 32       | sales_quotation_view     |
| 13 | POST   | /api/sale-quotations                                      | 32       | sales_quotation_create   |
| 14 | PUT    | /api/sale-quotations/{id}                                 | 32       | sales_quotation_edit     |
| 15 | PUT    | /api/sale-quotations/{id}/send                            | 32       | sales_quotation_send     |
| 16 | PUT    | /api/sale-quotations/{id}/lines/{lineId}/response         | 32       | sales_quotation_edit     |
| 17 | POST   | /api/sale-quotations/{id}/convert                         | 32       | sales_order_create       |
| 18 | GET    | /api/fulfillment-routes                                   | 33       | fulfillment_route_view   |
| 19 | GET    | /api/fulfillment-routes/{id}                              | 33       | fulfillment_route_view   |
| 20 | POST   | /api/fulfillment-routes                                   | 33       | fulfillment_route_manage |
| 21 | PUT    | /api/fulfillment-routes/{id}                              | 33       | fulfillment_route_manage |
| 22 | DELETE | /api/fulfillment-routes/{id}                              | 33       | fulfillment_route_manage |

| #  | Method  | Endpoint                                 | Addendum | Permission Required       |
|----|---------|------------------------------------------|----------|---------------------------|
| 23 | PUT     | /api/fulfillment-routes/{id}/set-default | 33       | fulfillment_route_manage  |
| 24 | POST    | /api/product-variants/bulk-assign-route  | 33       | fulfillment_route_assign  |
| 25 | GET     | /api/sale-orders/{id}/delivery-preview   | 33       | sales_order_view          |
| 26 | GET/PUT | /api/lead-time-defaults                  | 34       | lead_time_defaults_manage |

- All 26+ new endpoints exist and return correct responses
- Each endpoint has appropriate `[Authorize]` attribute
- Multi-tenancy filter (`org_id`) applied in all queries
- Pagination implemented on list endpoints
- Proper error responses (400, 404, 403)

## 3.2 Modified Endpoints

| Method   | Endpoint                      | Modification                                             | Addendum |
|----------|-------------------------------|----------------------------------------------------------|----------|
| POST/PUT | /api/sale-order-lines         | Min/Max qty validation                                   | 31       |
| GET      | /api/products/{id}            | Returns BOM inline, min/max fields                       | 31       |
| POST     | /api/sale-orders/{id}/confirm | Route resolution + split by category + confirmation gate | 33, 34   |
| GET      | /api/sale-orders/{id}         | Returns source linking, delivery indicators, route info  | 32, 33   |
| GET      | /api/product-variants/{id}    | Returns fulfillment_route_id, lead time columns          | 33, 34   |

- All modified endpoints reflect the new behavior
- Backward compatibility maintained for existing consumers

## PART 4 — UI COMPONENTS & WIREFRAMES

### 4.1 Addendum 31 UI (5 Wireframes)

| #  | Component                                    | Description                                                    | Verified |
|----|----------------------------------------------|----------------------------------------------------------------|----------|
| W1 | Product Form — Classification Tab            | Two sections: Product Classification + Manufacturing & Supply  | [ ]      |
| W2 | Product Form — BOM Panel                     | Two-panel layout: 35% BOM list + 65% BOM editor                | [ ]      |
| W3 | Production Order — Material Availability Tab | Available section + Shortage section + Reserve/Request buttons | [ ]      |
| W4 | Purchase Required Dashboard                  | Aggregated shortage list with affected POs drawer              | [ ]      |
| W5 | Allocation Run Dialog                        | Manual allocation trigger with warehouse/product filters       | [ ]      |

### 4.2 Addendum 32 UI (6 Wireframes)

| #  | Component                                 | Description                                                        | Verified |
|----|-------------------------------------------|--------------------------------------------------------------------|----------|
| W1 | Sale Inquiry List                         | Grid with status filter, customer filter, date range               | [ ]      |
| W2 | Sale Inquiry Form                         | Header + line grid with evaluation columns                         | [ ]      |
| W3 | Sale Quotation Form                       | Header + line grid with line_type, alternatives, customer response | [ ]      |
| W4 | Sale Order — Source & Customer PO Section | Source document link + customer PO fields                          | [ ]      |
| W5 | Sale Order — Delivery Indicators          | Colored dots on line grid + Reserve/Release buttons                | [ ]      |
| W6 | Rejection Reasons Management              | Settings page CRUD for rejection reasons                           | [ ]      |

### 4.3 Addendum 33 UI (4 Wireframes)

| #  | Component                          | Description                                                       | Verified |
|----|------------------------------------|-------------------------------------------------------------------|----------|
| W1 | Fulfillment Route Configuration    | Route list + route editor with step builder                       | [ ]      |
| W2 | Product Variant — Route Assignment | Route dropdown on variant form                                    | [ ]      |
| W3 | Sale Order Line — Route Selector   | Route dropdown per SO line + effective route display              | [ ]      |
| W4 | Delivery Preview Panel             | Grouped preview showing expected DOs by route before confirmation | [ ]      |

#### 4.4 Addendum 34 UI (5 Wireframes)

| #  | Component                             | Description                                       | Verified |
|----|---------------------------------------|---------------------------------------------------|----------|
| W1 | Route Configuration — Category Column | Category dropdown added to route create/edit      | [ ]      |
| W2 | Product Variant — Lead Time Section   | Conditional lead time fields by route type        | [ ]      |
| W3 | Lead Time Defaults Settings Page      | Org-level default management form                 | [ ]      |
| W4 | SO Confirmation — Split Summary       | Preview showing which lines create DOs vs POs     | [ ]      |
| W5 | Production Order — Delivery Link      | DO link on PO detail, shortfall notification area | [ ]      |

## PART 5 — BUSINESS RULES MASTER INDEX (107 Total)

### 5.1 Addendum 31 Business Rules (24)

| Rule ID   | Change | Summary                                                                | Verified |
|-----------|--------|------------------------------------------------------------------------|----------|
| BR-C1-01  | C1     | Min qty validation on SO line add/edit                                 | [ ]      |
| BR-C1-02  | C1     | Max qty validation on SO line add/edit                                 | [ ]      |
| BR-C1-03  | C1     | Validation at line level, not confirmation                             | [ ]      |
| BR-C1-04  | C1     | NULL means no limit                                                    | [ ]      |
| BR-C1-05  | C1     | Cross-validation: min ≤ max on Product save                            | [ ]      |
| BR-C1-06  | C1     | Min/Max displayed as helper text on SO line form                       | [ ]      |
| BR-C3-01  | C3     | Engine creates PO in DRAFT (not auto-submitted)                        | [ ]      |
| BR-C3-02  | C3     | Supplier priority: default → last used → NULL                          | [ ]      |
| BR-C3-03  | C3     | No supplier: PO created with NULL, flagged                             | [ ]      |
| BR-C3-04  | C3     | Notification to purchase_order_write users                             | [ ]      |
| BR-C3-05  | C3     | Draft PO includes Production Order reference                           | [ ]      |
| BR-C3-06  | C3     | Duplicate prevention for same product + PO                             | [ ]      |
| BR-C3-07  | C3     | Quantities match shortage amounts                                      | [ ]      |
| BR-C4-01  | C4     | BOM mandatory for manufacturing products                               | [ ]      |
| BR-C4-02  | C4     | BOM section visible only for manufacturing products                    | [ ]      |
| BR-C6-01  | C6     | PMR warehouse ALWAYS from ProductionOrder.production_warehouse_id      | [ ]      |
| BR-C7-01  | C7     | sale_order_id and sale_order_line_id populated on PO creation          | [ ]      |
| BR-C8-01  | C8     | Reserve calls IStockReservationService.Reserve()                       | [ ]      |
| BR-C8-02  | C8     | Allocation Request creates sourcing request                            | [ ]      |
| BR-C9-01  | C9     | Aggregates shortages across all active POs per product                 | [ ]      |
| BR-C9-02  | C9     | Duplicate PO prevention with "PO Pending" badge                        | [ ]      |
| BR-C10-01 | C10    | GRN → allocation is MANUAL                                             | [ ]      |
| BR-C10-02 | C10    | FGR → allocation is AUTOMATIC                                          | [ ]      |
| BR-C10-03 | C10    | Material status update after allocation (SHORTAGE → AVAILABLE/PARTIAL) | [ ]      |
| BR-C10-04 | C10    | Full allocation → READY status                                         | [ ]      |

### 5.2 Addendum 32 Business Rules (28)

| Rule ID  | Change | Summary                                             | Verified |
|----------|--------|-----------------------------------------------------|----------|
| BR-C1-01 | C1     | Inquiry number: INQ-{YYYY}-{NNNN}                   | [ ]      |
| BR-C1-02 | C1     | New inquiry starts RECEIVED                         | [ ]      |
| BR-C1-03 | C1     | UNDER_REVIEW requires assigned_to                   | [ ]      |
| BR-C1-04 | C1     | Line evaluation starts PENDING                      | [ ]      |
| BR-C1-05 | C1     | REVIEW_COMPLETE requires all lines non-PENDING      | [ ]      |
| BR-C1-06 | C1     | QUOTED creates linked Quotation                     | [ ]      |
| BR-C1-07 | C1     | DECLINED requires CANNOT_SUPPLY or rejection reason | [ ]      |

| Rule ID  | Change | Summary                                                        | Verified |
|----------|--------|----------------------------------------------------------------|----------|
| BR-C2-01 | C2     | Quotation number: SQ-{YYYY}-{NNNN}                             | [ ]      |
| BR-C2-02 | C2     | New quotation starts DRAFT                                     | [ ]      |
| BR-C2-03 | C2     | SENT requires $\geq 1$ NORMAL line with price > 0              | [ ]      |
| BR-C2-04 | C2     | SENT records sent_at timestamp                                 | [ ]      |
| BR-C2-05 | C2     | Customer response: PENDING/ACCEPTED/REJECTED/COUNTER           | [ ]      |
| BR-C2-06 | C2     | COUNTER requires counter_price or counter_qty                  | [ ]      |
| BR-C2-07 | C2     | ACCEPTED status requires $\geq 1$ ACCEPTED line                | [ ]      |
| BR-C2-08 | C2     | ALTERNATIVE lines via parent_line_id                           | [ ]      |
| BR-C2-09 | C2     | REJECTED lines require rejection_reason_id                     | [ ]      |
| BR-C2-10 | C2     | QuotationExpiryJob: SENT → EXPIRED when valid_until < today    | [ ]      |
| BR-C2-11 | C2     | CONVERTED creates SO with ACCEPTED lines only                  | [ ]      |
| BR-C3-01 | C3     | source_type auto-set by creation path                          | [ ]      |
| BR-C3-02 | C3     | source_quotation_id populated on conversion                    | [ ]      |
| BR-C3-03 | C3     | source_inquiry_id populated when traceable                     | [ ]      |
| BR-C3-04 | C3     | customer_po_reference max 50 chars                             | [ ]      |
| BR-C3-05 | C3     | customer_po_attachment_id links to attachment                  | [ ]      |
| BR-C4-01 | C4     | Delivery indicator computed at read time                       | [ ]      |
| BR-C4-02 | C4     | Reserve requires sales_reserve_inventory                       | [ ]      |
| BR-C4-03 | C4     | Release requires sales_release_reservation                     | [ ]      |
| BR-C4-04 | C4     | Reserve calls IStockReservationService with SALES_ORDER source | [ ]      |
| BR-C4-05 | C4     | Release calls IStockReservationService.Release()               | [ ]      |
| BR-C4-06 | C4     | reserved_qty updated on SaleOrderLines                         | [ ]      |
| BR-C4-07 | C4     | SO cancel auto-releases reservations                           | [ ]      |
| BR-C4-08 | C4     | Reserve qty $\leq$ (ordered - delivered)                       | [ ]      |

### 5.3 Addendum 33 Business Rules (25)

| Rule ID  | Change | Summary                                               | Verified |
|----------|--------|-------------------------------------------------------|----------|
| BR-C1-01 | C1     | Step codes: PICK/PACK/STAGE/APPROVAL/GOODS_ISSUE/SHIP | [ ]      |
| BR-C1-02 | C1     | Steps ordered by step_order                           | [ ]      |
| BR-C1-03 | C1     | One default route per org                             | [ ]      |
| BR-C1-04 | C1     | System routes cannot be deleted                       | [ ]      |
| BR-C1-05 | C1     | System routes can be deactivated                      | [ ]      |
| BR-C1-06 | C1     | Route code unique per org                             | [ ]      |
| BR-C1-07 | C1     | At least one step per route                           | [ ]      |
| BR-C1-08 | C1     | PICK step recommended (warning if absent)             | [ ]      |
| BR-C1-09 | C1     | Custom routes with any step combination               | [ ]      |
| BR-C2-01 | C2     | Variant route_id NULL = use org default               | [ ]      |
| BR-C2-02 | C2     | Bulk route assignment by category                     | [ ]      |
| BR-C3-01 | C3     | 4-level route resolution priority                     | [ ]      |
| BR-C3-02 | C3     | NULL route blocks confirmation                        | [ ]      |
| BR-C3-03 | C3     | Error identifies lines with no route                  | [ ]      |
| BR-C3-04 | C3     | Route dropdown shows active routes                    | [ ]      |

| Rule ID  | Change | Summary                                               | Verified |
|----------|--------|-------------------------------------------------------|----------|
| BR-C3-05 | C3     | Delivery Preview computed on-the-fly                  | [ ]      |
| BR-C4-01 | C4     | SO confirmation creates DOs grouped by route          | [ ]      |
| BR-C4-02 | C4     | One DO per route group                                | [ ]      |
| BR-C4-03 | C4     | DO created in DRAFT status                            | [ ]      |
| BR-C4-04 | C4     | DO from_source_type = SALE_ORDER                      | [ ]      |
| BR-C4-05 | C4     | DO sale_order_id and fulfillment_route_id populated   | [ ]      |
| BR-C4-06 | C4     | DO lines link via sale_order_line_id                  | [ ]      |
| BR-C4-07 | C4     | Cancel SO → cancel DOs not yet GOODS_ISSUED           | [ ]      |
| BR-C5-01 | C5     | State machine skips statuses not in route             | [ ]      |
| BR-C5-02 | C5     | Skipping is transparent (direct transition)           | [ ]      |
| BR-C5-03 | C5     | PICK_ONLY: skips PACK, STAGE, APPROVAL, SHIP statuses | [ ]      |
| BR-C5-04 | C5     | All 15 statuses remain in enum                        | [ ]      |
| BR-C5-05 | C5     | Admin manual override still available                 | [ ]      |

## 5.4 Addendum 34 Business Rules (30)

| Rule ID  | Change | Summary                                                                        | Verified |
|----------|--------|--------------------------------------------------------------------------------|----------|
| BR-C1-01 | C1     | route_category: STOCK/MANUFACTURE/BUY/DROPSHIP                                 | [ ]      |
| BR-C1-02 | C1     | Existing routes → STOCK                                                        | [ ]      |
| BR-C1-03 | C1     | New manufacturing routes → MANUFACTURE                                         | [ ]      |
| BR-C1-04 | C1     | Category displayed and user-selectable for custom routes                       | [ ]      |
| BR-C2-01 | C2     | BOM visible when route_category = MANUFACTURE                                  | [ ]      |
| BR-C2-02 | C2     | is_manufacturable auto-synced from route                                       | [ ]      |
| BR-C2-03 | C2     | Legacy boolean filters preserved                                               | [ ]      |
| BR-C2-04 | C2     | Route change hides BOM tab, preserves data                                     | [ ]      |
| BR-C3-01 | C3     | Variant overrides org defaults (NULL = use org)                                | [ ]      |
| BR-C3-02 | C3     | One LeadTimeDefaults row per org                                               | [ ]      |
| BR-C3-03 | C3     | Field visibility by route type                                                 | [ ]      |
| BR-C3-04 | C3     | MANUFACTURE shows: mfg, mfg_buffer, quality, pick_pack, shipping, sales_buffer | [ ]      |
| BR-C3-05 | C3     | STOCK shows: supplier, quality, pick_pack, shipping, sales_buffer              | [ ]      |
| BR-C4-01 | C4     | BOM traversal: sub-components parallel (MAX)                                   | [ ]      |
| BR-C4-02 | C4     | This-level operations sequential (SUM)                                         | [ ]      |
| BR-C4-03 | C4     | Stock-aware: stock available → lead = 0                                        | [ ]      |
| BR-C4-04 | C4     | Cycle detection → return 0 + log warning                                       | [ ]      |
| BR-C4-05 | C4     | 5-min cache per (variant_id, qty_bucket)                                       | [ ]      |
| BR-C4-06 | C4     | BOM operations: 480 min = 1 working day                                        | [ ]      |
| BR-C5-01 | C5     | STOCK lines → SaleOrderDeliveryCreator                                         | [ ]      |
| BR-C5-02 | C5     | MANUFACTURE lines → SaleOrderProductionCreator                                 | [ ]      |
| BR-C5-03 | C5     | Mixed orders process both groups                                               | [ ]      |
| BR-C5-04 | C5     | One PO per manufacturing line                                                  | [ ]      |
| BR-C5-05 | C5     | PO includes: sale_order_id, sale_order_line_id, fulfillment_route_id           | [ ]      |
| BR-C5-06 | C5     | BOM explosion triggered on PO creation                                         | [ ]      |

| Rule ID  | Change | Summary                                                         | Verified |
|----------|--------|-----------------------------------------------------------------|----------|
| BR-C6-01 | C6     | FGR triggers automatic allocation (existing)                    | [ ]      |
| BR-C6-02 | C6     | FGR auto-creates DO if PO has SO link + route                   | [ ]      |
| BR-C6-03 | C6     | delivery_qty = actual_qty (actual yield)                        | [ ]      |
| BR-C6-04 | C6     | Shortfall: create delivery for actual + notify                  | [ ]      |
| BR-C6-05 | C6     | Zero production: no delivery + error notification               | [ ]      |
| BR-C6-06 | C6     | PO.delivery_order_id populated                                  | [ ]      |
| BR-C6-07 | C6     | DO from_source_type extended for production source              | [ ]      |
| BR-C6-08 | C6     | MTS escape hatch: stock route override skips production         | [ ]      |
| BR-C6-09 | C6     | Shortfall notification: PO number, planned vs actual, shortfall | [ ]      |

## PART 6 — TEST SCENARIOS MASTER INDEX (128 Total)

### 6.1 Addendum 31 Test Scenarios (29)

| #  | Scenario                                 | Expected Result                                      | Verified |
|----|------------------------------------------|------------------------------------------------------|----------|
| 1  | Add SO line with qty < product min_qty   | Blocked with validation message                      | [ ]      |
| 2  | Add SO line with qty > product max_qty   | Blocked with validation message                      | [ ]      |
| 3  | Add SO line with qty within range        | Accepted                                             | [ ]      |
| 4  | Add SO line with NULL min/max on product | Accepted (no limit)                                  | [ ]      |
| 5  | Save product with min > max              | Blocked with cross-validation error                  | [ ]      |
| 6  | View Product form — Classification tab   | Shows both sections (Classification + Manufacturing) | [ ]      |
| 7  | Supply Requirement Engine triggered      | Draft PO created (not submitted)                     | [ ]      |
| 8  | Product has default supplier             | Draft PO uses default supplier                       | [ ]      |
| 9  | Product has no default but has past POs  | Draft PO uses last supplier                          | [ ]      |
| 10 | Product has no supplier history          | Draft PO created with NULL supplier                  | [ ]      |
| 11 | Draft PO already exists for product+PO   | No duplicate created                                 | [ ]      |
| 12 | Notification sent to purchase team       | Users with purchase_order_write receive notification | [ ]      |
| 13 | View BOM on Product form                 | Two-panel layout, BOM list + editor                  | [ ]      |

| #  | Scenario                                       | Expected Result                                               | Verified |
|----|------------------------------------------------|---------------------------------------------------------------|----------|
| 14 | Manufacturing product without BOM              | Save blocked, BOM required                                    | [ ]      |
| 15 | Create new BOM                                 | effective_from defaults to today                              | [ ]      |
| 16 | BOM line form                                  | No warehouse dropdown present                                 | [ ]      |
| 17 | PMR creation                                   | Warehouse from PO.production_warehouse_id                     | [ ]      |
| 18 | PO created from SO confirmation                | sale_order_id and sale_order_line_id set                      | [ ]      |
| 19 | Material Availability — all available          | "Available" section populated, Reserve button shown           | [ ]      |
| 20 | Material Availability — shortage               | "Shortage" section populated, Allocation button shown         | [ ]      |
| 21 | Reserve All Available                          | All available materials reserved via IStockReservationService | [ ]      |
| 22 | Purchase Required dashboard                    | Aggregated shortages grouped by product                       | [ ]      |
| 23 | Create PO from dashboard (no duplicate exists) | Draft PO created                                              | [ ]      |
| 24 | Create PO from dashboard (duplicate exists)    | "PO Pending" badge shown, creation blocked                    | [ ]      |
| 25 | GRN received                                   | Manual allocation available via /api/allocation/run           | [ ]      |
| 26 | Run manual allocation                          | Materials allocated, PO material status updated               | [ ]      |
| 27 | FGR confirmed                                  | Automatic allocation triggered by event handler               | [ ]      |
| 28 | Full allocation                                | PO material status → READY                                    | [ ]      |
| 29 | Partial allocation                             | PO material status → PARTIAL                                  | [ ]      |

## 6.2 Addendum 32 Test Scenarios (34)

| # | Scenario            | Expected Result                                       | Verified |
|---|---------------------|-------------------------------------------------------|----------|
| 1 | Create Sale Inquiry | Inquiry created with status RECEIVED, number INQ-xxxx | [ ]      |

| #  | Scenario                                             | Expected Result                                     | Verified |
|----|------------------------------------------------------|-----------------------------------------------------|----------|
| 2  | Transition to UNDER_REVIEW without assigned_to       | Blocked                                             | [ ]      |
| 3  | Transition to UNDER_REVIEW with assigned_to          | Succeeds                                            | [ ]      |
| 4  | Evaluate line as CAN_SUPPLY                          | Line evaluation updated                             | [ ]      |
| 5  | Evaluate line as CAN_NOT_SUPPLY                      | Rejection reason required                           | [ ]      |
| 6  | Transition to REVIEW_COMPLETE with PENDING lines     | Blocked                                             | [ ]      |
| 7  | Transition to REVIEW_COMPLETE, all evaluated         | Succeeds                                            | [ ]      |
| 8  | Transition to QUOTED                                 | Linked quotation created                            | [ ]      |
| 9  | Create Sale Quotation                                | Quotation created with status DRAFT, number SQ-xxxx | [ ]      |
| 10 | Add ALTER_NATIVE line                                | parent_line_id set, line_type = ALTERNATIVE         | [ ]      |
| 11 | Transition to SENT without valid lines               | Blocked (needs ≥1 NORMAL with price)                | [ ]      |
| 12 | Transition to SENT with valid lines                  | Succeeds, sent_at recorded                          | [ ]      |
| 13 | Set customer response COUNTER without counter values | Blocked                                             | [ ]      |
| 14 | Set customer response COUNTER with counter_price     | Succeeds                                            | [ ]      |
| 15 | Set customer response ACCEPTED                       | Line marked accepted                                | [ ]      |

| #  | Scenario                                                    | Expected Result                                                   | Verified |
|----|-------------------------------------------------------------|-------------------------------------------------------------------|----------|
| 16 | Transition to ACCEPTED without any accepted lines           | Blocked                                                           | [ ]      |
| 17 | Transition to ACCEPTED with accepted lines                  | Succeeds                                                          | [ ]      |
| 18 | QuotationExpiryJob — SENT quotation past valid_until        | Status → EXPIRED                                                  | [ ]      |
| 19 | QuotationExpiryJob — DRAFT quotation past valid_until       | Not affected (stays DRAFT)                                        | [ ]      |
| 20 | Convert quotation to SO                                     | SO created with ACCEPTED lines only, source_type = FROM_QUOTATION | [ ]      |
| 21 | Convert quotation — COUNTER lines without seller acceptance | Not included in SO                                                | [ ]      |
| 22 | Quotation after conversion                                  | Status = CONVERTED, read-only                                     | [ ]      |
| 23 | Create SO manually                                          | source_type = MANUAL, no source links                             | [ ]      |
| 24 | View SO with source quotation                               | Source document link clickable                                    | [ ]      |
| 25 | Set customer PO reference                                   | Saved to customer_po_reference field                              | [ ]      |
| 26 | Delivery indicator — fully fulfilled line                   | GREEN dot                                                         | [ ]      |
| 27 | Delivery indicator — fully reserved, not fulfilled          | BLUE dot                                                          | [ ]      |
| 28 | Delivery indicator — partially reserved                     | YELLOW dot                                                        | [ ]      |

| #  | Scenario                                           | Expected Result                             | Verified |
|----|----------------------------------------------------|---------------------------------------------|----------|
| 29 | Delivery indicator — nothing reserved or fulfilled | RED dot                                     | [ ]      |
| 30 | Delivery indicator — cancelled line                | GREY dot                                    | [ ]      |
| 31 | Reserve inventory on SO line                       | reserved_qty updated, indicator changes     | [ ]      |
| 32 | Release reservation                                | reserved_qty decremented, indicator changes | [ ]      |
| 33 | Cancel SO with reservations                        | All reservations auto-released              | [ ]      |
| 34 | Reserve without permission                         | 403 Forbidden                               | [ ]      |

### 6.3 Addendum 33 Test Scenarios (30)

| #  | Scenario                         | Expected Result                                               | Verified |
|----|----------------------------------|---------------------------------------------------------------|----------|
| 1  | View seed routes                 | 3 routes displayed (PICK_ONLY, PICK_AND_SHIP, PICK_PACK_SHIP) | [ ]      |
| 2  | Create custom route              | Route created with specified steps                            | [ ]      |
| 3  | Delete system route              | Blocked (is_system protection)                                | [ ]      |
| 4  | Deactivate system route          | Succeeds (is_active = 0)                                      | [ ]      |
| 5  | Set default route                | Previous default unset, new default set                       | [ ]      |
| 6  | Create route without steps       | Blocked (at least 1 step required)                            | [ ]      |
| 7  | Create route with duplicate code | Blocked (unique constraint)                                   | [ ]      |
| 8  | Assign route to product variant  | fulfillment_route_id set                                      | [ ]      |
| 9  | Bulk assign route to category    | All variants in category updated                              | [ ]      |
| 10 | Variant with NULL route          | Resolves to org default                                       | [ ]      |
| 11 | SO line with explicit route      | Uses line-level route (priority 1)                            | [ ]      |

| #  | Scenario                                     | Expected Result                                                                    | Verified |
|----|----------------------------------------------|------------------------------------------------------------------------------------|----------|
| 12 | SO line without route, variant has route     | Uses variant route (priority 2)                                                    | [ ]      |
| 13 | SO line without route, variant without route | Uses org default (priority 3)                                                      | [ ]      |
| 14 | SO line resolves to NULL (no default)        | Confirmation blocked with error                                                    | [ ]      |
| 15 | Confirmation error message                   | Lists specific lines with no route                                                 | [ ]      |
| 16 | Delivery Preview                             | Shows grouped DOs by route (computed, not saved)                                   | [ ]      |
| 17 | Confirm SO — all lines same route            | 1 Delivery Order created                                                           | [ ]      |
| 18 | Confirm SO — lines with 2 different routes   | 2 Delivery Orders created                                                          | [ ]      |
| 19 | Confirm SO — lines with 3 different routes   | 3 Delivery Orders created                                                          | [ ]      |
| 20 | DO created                                   | Status = DRAFT, from_source_type = SALE_ORDER                                      | [ ]      |
| 21 | DO links                                     | sale_order_id, fulfillment_route_id populated                                      | [ ]      |
| 22 | DO lines                                     | sale_order_line_id populated                                                       | [ ]      |
| 23 | Cancel SO — DOs in DRAFT                     | DOs auto-cancelled                                                                 | [ ]      |
| 24 | Cancel SO — DO in PICKING                    | DO auto-cancelled                                                                  | [ ]      |
| 25 | Cancel SO — DO is GOODS_ISSUED               | Error: manual reversal required                                                    | [ ]      |
| 26 | PICK_ONLY route state machine                | RELEASED → PICKING → PICKED → GOODS_ISSUED → COMPLETED                             | [ ]      |
| 27 | PICK_AND_SHIP route state machine            | RELEASED → PICKING → PICKED → GOODS_ISSUED → IN_TRANSIT → ... → DELIVERED          | [ ]      |
| 28 | PICK_PACK_SHIP route state machine           | RELEASED → PICKING → PICKED → PACKED → GOODS_ISSUED → IN_TRANSIT → ... → DELIVERED | [ ]      |

| #  | Scenario                     | Expected Result                                 | Verified |
|----|------------------------------|-------------------------------------------------|----------|
| 29 | Status skip — no PACK step   | Transition from PICKED directly to GOODS_ISSUED | [ ]      |
| 30 | Admin manual status override | Override works regardless of route              | [ ]      |

## 6.4 Addendum 34 Test Scenarios (35)

| #  | Scenario                                       | Expected Result                                                     | Verified |
|----|------------------------------------------------|---------------------------------------------------------------------|----------|
| 1  | View routes — category column                  | STOCK/MANUFACTURE labels displayed                                  | [ ]      |
| 2  | Create custom route with MANUFACTURE category  | Route created with correct category                                 | [ ]      |
| 3  | Seed routes category values                    | PICK_ONLY/PICK_AND_SHIP/PICK_PACK_SHIP = STOCK; MFG_* = MANUFACTURE | [ ]      |
| 4  | New org creation                               | Both STOCK and MANUFACTURE routes seeded                            | [ ]      |
| 5  | Variant with MANUFACTURE route                 | BOM tab visible                                                     | [ ]      |
| 6  | Variant with STOCK route                       | BOM tab hidden                                                      | [ ]      |
| 7  | Change variant route from MANUFACTURE to STOCK | BOM tab hidden, existing BOMs preserved                             | [ ]      |
| 8  | is_manufacturable sync                         | TRUE when route = MANUFACTURE, FALSE otherwise                      | [ ]      |
| 9  | Legacy filter is_manufacturable=true           | Returns variants with MANUFACTURE route                             | [ ]      |
| 10 | Lead Time Defaults — view                      | Shows org defaults with current values                              | [ ]      |
| 11 | Lead Time Defaults — update                    | Values saved, requires lead_time_defaults_manage                    | [ ]      |
| 12 | Variant lead time — MANUFACTURE route          | Shows mfg fields, hides supplier field                              | [ ]      |

| #  | Scenario                                      | Expected Result                                                  | Verified |
|----|-----------------------------------------------|------------------------------------------------------------------|----------|
| 13 | Variant lead time — STOCK route               | Shows supplier field, hides mfg fields                           | [ ]      |
| 14 | Lead time resolution — variant override set   | Uses variant value                                               | [ ]      |
| 15 | Lead time resolution — variant NULL           | Falls back to org default                                        | [ ]      |
| 16 | Calculate lead time — simple product (no BOM) | Sum of applicable components                                     | [ ]      |
| 17 | Calculate lead time — product with BOM        | Recursive: MAX(sub-components) + SUM(this-level)                 | [ ]      |
| 18 | Calculate lead time — sub-component has stock | Sub-component lead = 0                                           | [ ]      |
| 19 | Calculate lead time — BOM cycle               | Returns 0, logs warning                                          | [ ]      |
| 20 | Calculate lead time — caching                 | Second call within 5 min returns cached result                   | [ ]      |
| 21 | Inquiry line — lead time calculated           | calculated_lead_time_days and calculated_delivery_date populated | [ ]      |
| 22 | Quotation line — lead time calculated         | Same columns populated                                           | [ ]      |
| 23 | SO line — lead time calculated                | Same columns populated                                           | [ ]      |
| 24 | effective_delivery_date — no manual override  | = calculated_delivery_date                                       | [ ]      |
| 25 | effective_delivery_date — manual override set | = manual_delivery_date                                           | [ ]      |
| 26 | Confirm SO — all STOCK lines                  | Only DOs created (existing flow)                                 | [ ]      |
| 27 | Confirm SO — all MANUFACTURE lines            | Only POs created (new flow)                                      | [ ]      |

| #  | Scenario                                   | Expected Result                                                   | Verified |
|----|--------------------------------------------|-------------------------------------------------------------------|----------|
| 28 | Confirm SO — mixed STOCK + MANUFACTURE     | Both DOs and POs created                                          | [ ]      |
| 29 | PO from SO confirmation                    | sale_order_id, sale_order_line_id, fulfillment_route_id populated | [ ]      |
| 30 | PO BOM explosion                           | PMR lines created from active BOM                                 | [ ]      |
| 31 | PO backward scheduling                     | planned_start_date = effective_delivery_date - lead_time_days     | [ ]      |
| 32 | FGR confirmed — PO with SO link            | Delivery Order auto-created                                       | [ ]      |
| 33 | FGR — actual_qty < planned_qty             | DO created for actual_qty + shortfall notification                | [ ]      |
| 34 | FGR — actual_qty = 0                       | No DO created + error notification                                | [ ]      |
| 35 | MTS escape hatch — route override to STOCK | DO created immediately from stock, no PO                          | [ ]      |

## PART 7 — HANGFIRE JOBS

| Job                  | Schedule                             | Module | Addendum     | Verified |
|----------------------|--------------------------------------|--------|--------------|----------|
| QuotationExpiryJob   | Daily at 01:00 UTC                   | demand | 32           | [ ]      |
| ReservationExpiryJob | (Existing — verify still functional) | demand | Pre-existing | [ ]      |

- `QuotationExpiryJob` registered in Hangfire configuration
- Job runs at 01:00 UTC daily
- Job processes only SENT quotations (not DRAFT)
- Job updates status to EXPIRED where `valid_until < GETUTCDATE()`
- Job logs execution details
- `ReservationExpiryJob` still runs correctly after schema changes

## PART 8 — STATE MACHINES SUMMARY

### 8.1 Sale Inquiry State Machine (Addendum 32)

RECEIVED → UNDER\_REVIEW → REVIEW\_COMPLETE → QUOTED  
→ DECLINED

- All transitions implemented
- UNDER\_REVIEW requires assigned\_to
- REVIEW\_COMPLETE requires all lines evaluated (non-PENDING)
- QUOTED creates linked quotation
- DECLINED requires rejection justification
- Invalid transitions rejected with appropriate error

### 8.2 Sale Quotation State Machine (Addendum 32)

DRAFT → SENT → ACCEPTED → CONVERTED  
→ REJECTED  
→ EXPIRED (automatic via job)

- All transitions implemented
- SENT requires valid lines
- ACCEPTED requires accepted customer responses
- EXPIRED set by Hangfire job
- CONVERTED creates linked SO
- CONVERTED makes quotation read-only

### 8.3 Delivery Order State Machine — Route-Aware (Addendum 33)

#### Full 15-status chain:

DRAFT → RELEASED → PICKING → PICKED → PACKED → STAGED →  
PENDING\_APPROVAL → GOODS\_ISSUED → IN\_TRANSIT → AT\_HUB →  
OUT\_FOR\_DELIVERY → DELIVERED → COMPLETED  
(+ CANCELLED, RETURNED)

#### Step-to-status mapping:

| Step        | Statuses                                        |
|-------------|-------------------------------------------------|
| PICK        | RELEASED, PICKING, PICKED                       |
| PACK        | PACKED                                          |
| STAGE       | STAGED                                          |
| APPROVAL    | PENDING_APPROVAL                                |
| GOODS_ISSUE | GOODS_ISSUED                                    |
| SHIP        | IN_TRANSIT, AT_HUB, OUT_FOR_DELIVERY, DELIVERED |

#### Route examples:

| Route         | Active Statuses                                                                               |
|---------------|-----------------------------------------------------------------------------------------------|
| PICK_ONLY     | DRAFT → RELEASED → PICKING → PICKED → GOODS_ISSUED → COMPLETED                                |
| PICK_AND_SHIP | DRAFT → RELEASED → PICKING → PICKED → GOODS_ISSUED → IN_TRANSIT → ... → DELIVERED → COMPLETED |

| Route          | Active Statuses                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------|
| PICK_PACK_SHIP | DRAFT → RELEASED → PICKING → PICKED → PACKED →<br>GOODS_ISSUED → IN_TRANSIT → ... → DELIVERED →<br>COMPLETED |

- `RouteAwareDeliveryStateMachine` handles all route permutations
- Skipped statuses never appear in UI or API for that delivery
- `GetNextStatus()` and `GetPreviousStatus()` respect route steps

## PART 9 — KEY SERVICES REGISTRY

| Service/Interface                 | Module    | Addendum | Purpose                                 |
|-----------------------------------|-----------|----------|-----------------------------------------|
| SaleOrderLineService (validation) | demand    | 31       | Min/Max qty validation                  |
| SupplyRequirementEngine           | material  | 31       | Draft PO creation from shortages        |
| IMaterialReadinessService         | material  | 31       | Material availability queries           |
| IAllocationEngine                 | material  | 31       | Manual & automatic allocation           |
| SaleInquiryService                | demand    | 32       | Inquiry CRUD + state transitions        |
| SaleQuotationService              | demand    | 32       | Quotation CRUD + state + conversion     |
| QuotationExpiryJob                | demand    | 32       | Hangfire quotation expiry               |
| IStockReservationService          | inventory | 32       | Reserve/Release (existing, extended)    |
| FulfillmentRouteService           | logistics | 33       | Route CRUD                              |
| EffectiveRouteResolver            | logistics | 33       | 4-level route resolution                |
| SaleOrderDeliveryCreator          | logistics | 33       | DO creation from SO (STOCK lines)       |
| SaleOrderDeliveryPreviewService   | logistics | 33       | Preview (computed, not persisted)       |
| RouteAwareDeliveryStateMachine    | logistics | 33       | Route-aware status transitions          |
| ILeadTimeCalculator               | material  | 34       | Recursive BOM lead time calculation     |
| SaleOrderProductionCreator        | material  | 34       | PO creation from SO (MANUFACTURE lines) |
| ProductionDeliveryCreator         | logistics | 34       | DO creation from completed PO           |
| FGRConfirmedEventHandler          | material  | 31, 34   | Extended: allocation + delivery bridge  |
| SaleOrderService.ConfirmAsync     | demand    | 33, 34   | Modified: route gate + category split   |

## PART 10 — DEVELOPMENT PHASES & TIMELINE

### 10.1 Addendum 31 Phases (23 days sequential, ~18 parallel)

| Phase                     | Tasks                                            | Days | Dependencies |
|---------------------------|--------------------------------------------------|------|--------------|
| A — Schema                | M1-M3 migrations                                 | 3    | None         |
| B — Product UI            | Tab consolidation, BOM inline, min/max fields    | 5    | A            |
| C — Workflow              | Draft PO flow, effective date, warehouse removal | 3    | A            |
| D — Material Availability | Tab UI, reserve action, allocation request       | 5    | B, C         |
| E — Purchase Required     | Dashboard, aggregation, duplicate prevention     | 4    | D            |
| F — Testing               | All 29 test scenarios                            | 3    | E            |

### 10.2 Addendum 32 Phases (23 days)

| Phase             | Tasks                                    | Days | Dependencies        |
|-------------------|------------------------------------------|------|---------------------|
| A — Schema        | M1-M7 migrations                         | 2    | None                |
| B — Inquiry       | Service, UI, state machine               | 5    | A                   |
| C — Quotation     | Service, UI, state machine, Hangfire job | 6    | A (parallel with B) |
| D — SO Extensions | Source linking, customer PO, attachments | 3    | B, C                |
| E — Indicators    | Delivery indicators, reservation RBAC    | 4    | D                   |

| Phase       | Tasks                 | Days | Dependencies |
|-------------|-----------------------|------|--------------|
| F — Testing | All 34 test scenarios | 3    | E            |

### 10.3 Addendum 33 Phases (19 days)

| Phase                  | Tasks                                | Days | Dependencies |
|------------------------|--------------------------------------|------|--------------|
| A — Route Config       | Tables, seed data, CRUD UI           | 3    | None         |
| B — Variant Assignment | Column, dropdown, bulk assign        | 2    | A            |
| C — SO Route           | Resolver, confirmation gate, preview | 4    | A, B         |
| D — Delivery Creation  | SaleOrderDeliveryCreator, DO linking | 4    | C            |
| E — State Machine      | RouteAwareDeliveryStateMachine       | 3    | A, D         |
| F — Testing            | All 30 test scenarios                | 3    | E            |

### 10.4 Addendum 34 Phases (25 days sequential, ~16 parallel)

| Phase              | Tasks                                               | Days | Dependencies        |
|--------------------|-----------------------------------------------------|------|---------------------|
| A — Route Category | Column, manufacturing seeds, cursor migration       | 4    | Add-33 complete     |
| B — Lead Time      | Defaults table, variant columns, settings UI        | 4    | A (parallel with D) |
| C — Calculation    | ILeadTime Calculator, BOM traversal, caching        | 5    | B                   |
| D — SO Split       | ConfirmAsync modification, ProductionCreator        | 5    | A (parallel with B) |
| E — Bridge         | FGRConfirmedEventHandler, ProductionDeliveryCreator | 4    | C, D                |
| F — Testing        | All 35 test scenarios                               | 3    | E                   |

## 10.5 Consolidated Timeline

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**Sequential (worst case): 90 dev days**

**With parallelization: ~76 dev days**

**Parallel execution opportunities:**

- Addendum 31 and 32 can start simultaneously (independent)
- Within Addendum 32: Inquiry (Phase B) and Quotation (Phase C) can run in parallel
- Within Addendum 34: Lead Time (Phase B) and SO Split (Phase D) can run in parallel
- Within Addendum 34: Calculation (Phase C) and Bridge (Phase E) overlap potential

**Critical path:** Addendum 33 → Addendum 34 (strict dependency on route infrastructure)

---

## PART 11 — CROSS-ADDENDUM INTEGRATION POINTS

These are the points where implementations from different addendums interact. Verify these connections specifically.

### 11.1 Production Order → Sale Order Link (Add-31 M2 + Add-34 C5)

- Add-31 creates the FK columns on ProductionOrders
- Add-34 C5 `SaleOrderProductionCreator` populates these columns
- Both addendums reference the same columns: `sale_order_id`, `sale_order_line_id`

### 11.2 Fulfillment Route → Route Category (Add-33 M1 + Add-34 M1)

- Add-33 creates the `FulfillmentRoutes` table
- Add-34 adds `route_category` to the same table
- Seed routes from Add-33 get `route_category = 'STOCK'` in Add-34
- Add-34 adds manufacturing seed routes to the same table

### 11.3 SO Confirmation Flow (Add-33 C4 + Add-34 C5)

- Add-33 introduces `SaleOrderDeliveryCreator` for all SO lines
- Add-34 modifies `ConfirmAsync` to split: STOCK → `SaleOrderDeliveryCreator`, MANUFACTURE → `SaleOrderProductionCreator`
- The delivery creator from Add-33 is reused unchanged for STOCK lines

### 11.4 FGRConfirmedEventHandler (Add-31 C10 + Add-34 C6)

- Add-31 establishes automatic allocation on FGR
- Add-34 extends the same handler to also create Delivery Orders
- Both behaviors must co-exist: allocation first, then delivery creation
- Handler checks for `sale_order_id` AND `fulfillment_route_id` before creating DO

### 11.5 IStockReservationService (Add-32 C4 + Add-33)

- Add-32 adds SALES\_ORDER as a reservation source type
- Add-33 introduces route-based delivery (may consume reservations)
- SO cancellation auto-releases reservations (Add-32) AND auto-cancels DOs (Add-33)

### 11.6 Sale Order Lines Accumulated Columns

Final state of `demand.SaleOrderLines` after all addendums:

| Column                    | Type                                | Addendum | Purpose              |
|---------------------------|-------------------------------------|----------|----------------------|
| <code>reserved_qty</code> | DECIMAL(18,4) NOT NULL<br>DEFAULT 0 | 32       | Reservation tracking |

| Column                                 | Type               | Addendum | Purpose                      |
|----------------------------------------|--------------------|----------|------------------------------|
| <code>fulfillment_route_id</code>      | INT NULL FK        | 33       | Route assignment             |
| <code>calculated_lead_time_days</code> | INT NULL           | 34       | Calculated lead time         |
| <code>calculated_delivery_date</code>  | DATE NULL          | 34       | Calculated delivery          |
| <code>manual_delivery_date</code>      | DATE NULL          | 34       | Manual override              |
| <code>lead_time_calculated_at</code>   | DATETIME 2 NULL    | 34       | Calculation timestamp        |
| <code>effective_delivery_date</code>   | COMPUTED PERSISTED | 34       | COALESCE(manual, calculated) |

- All columns present on `demand.SaleOrderLines`
- No naming conflicts
- EF Core entity has all properties mapped

## 11.7 Production Order Accumulated Columns

Final state of `material.ProductionOrders` after all addendums:

| Column                            | Type           | Addendum | Purpose                     |
|-----------------------------------|----------------|----------|-----------------------------|
| <code>sale_order_id</code>        | BIGINT NULL FK | 31       | SO traceability             |
| <code>sale_order_line_id</code>   | BIGINT NULL FK | 31       | SO line traceability        |
| <code>fulfillment_route_id</code> | INT NULL FK    | 34       | Route for delivery creation |
| <code>delivery_order_id</code>    | BIGINT NULL FK | 34       | Created DO traceability     |

- All columns present on `material.ProductionOrders`
- All FK constraints valid
- All filtered indexes created

## 11.8 Delivery Order Source Types

After all addendums, `from_source_type` on `logistics.delivery_orders` must support:

| Source Type             | Origin                         | Addendum |
|-------------------------|--------------------------------|----------|
| (existing values)       | Pre-existing flows             | Pre-31   |
| <code>SALE_ORDER</code> | SO confirmation (STOCK lines)  | 33       |
| Production source       | FGR bridge (MANUFACTURE lines) | 34       |

- CHECK constraint updated to include all source types
- Each creation path sets the correct source type

---

## PART 12 — FINAL VERIFICATION CHECKLIST

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### Schema Verification

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- All 18 migrations applied successfully in sequence
- No orphaned FK constraints
- All indexes created
- All seed data inserted for all existing orgs
- New org creation seeds all lookup data (rejection reasons, routes, lead time defaults)

### Backend Verification

---

- All 18 services/handlers listed in Part 9 implemented
- All 107 business rules verified (Parts 5.1–5.4)
- All state machines function correctly (Part 8)
- All Hangfire jobs registered and scheduled (Part 7)
- Multi-tenancy (org\_id filter) applied everywhere
- Audit columns (created\_by, modified\_by, timestamps) populated

### API Verification

---

- All 26+ new endpoints functional (Part 3.1)
- All modified endpoints updated (Part 3.2)
- Authorization attributes applied
- Pagination on list endpoints
- Proper error responses

### UI Verification

---

- All 20 UI wireframes/components implemented (Part 4)
- Responsive design
- Permission-based visibility (buttons hidden when no permission)
- Navigation between linked entities (Inquiry → Quotation → SO → DO/PO)

### Integration Verification

---

- All 8 cross-addendum integration points verified (Part 11)
- End-to-end flow: Inquiry → Quotation → SO Confirmation → Production/Delivery
- End-to-end flow: Production Complete → FGR → Allocation → Delivery Creation

### Testing Verification

---

- All 128 test scenarios pass (Part 6)
- Addendum 31: 29 scenarios ✓

- Addendum 32: 34 scenarios ✓
  - Addendum 33: 30 scenarios ✓
  - Addendum 34: 35 scenarios ✓
- 

**Document End**

\*SMS Consolidated Implementation Verification — Addendums 31, 32, 33 & 34\*

\*Generated: 2026-10-06\*